POLYTECHNIQUE MONTRÉAL
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Numerical Modeling of Multilayer Aircraft Ice Accretion

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DEDICATION

À Mama, Baba et Lilly
je vous aime . . .

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RÉSUMÉ

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ABSTRACT

Texte / Text.

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IETF	Internet Engineering Task Force
OSI	Open Systems Interconnection

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CHAPTER 1 INTRODUCTION

This chapter introduces the context of aircraft icing, its impact on flight safety, and its role in certification. It reviews the main ice morphologies, regulatory requirements, and certification tools, before presenting the challenges of numerical ice accretion modeling. The chapter concludes by defining the thesis objectives and outlining the organization of the thesis.

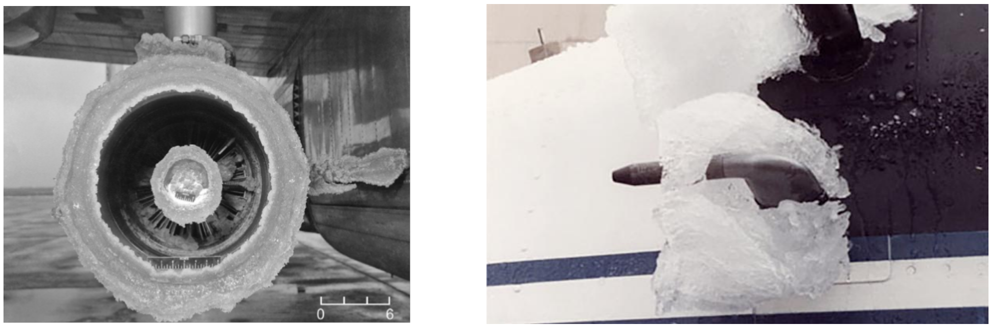
1.1 Context

In the context of climate change and increasingly stringent environmental regulations, the aeronautical industry is moving toward the development of more efficient and lower-emission aircraft. This transition introduces new aerodynamic, structural, and operational challenges, while maintaining flight safety as the primary requirement. It also increases the need for reliable certification procedures capable of assessing aircraft performance and safety over a wide range of operating conditions.

Among the phenomena considered during aircraft certification, in-flight icing remains one of the major concerns [6]. Ice accretion occurs when an aircraft flies through clouds containing supercooled water droplets[1]. When these droplets impinge on exposed surfaces and the local thermal conditions allow freezing, ice can accumulate on the aircraft geometry. The resulting ice shapes may significantly modify the aerodynamic configuration of the aircraft, leading to increased drag, reduced lift, altered stall behavior, and degraded controllability [6–9]. While aircraft icing is commonly associated with large exposed components such as wings, empennages, the aircraft nose, and engines, it may also affect smaller but critical components, including probes and sensors, as illustrated in Figure 1.1. Ice accretion on these components can produce erroneous measurements and, consequently, compromise flight safety.

Aircraft icing was not initially regarded as a primary concern during the early years of powered flight. Because aircraft were not yet equipped for routine instrument flight, pilots generally avoided clouds, and encounters with icing conditions were mostly inadvertent. This situation changed in the mid-1920s, when the U.S. Air Mail Service began operating scheduled day-and-night routes between New York and Chicago. These operations exposed pilots to icing conditions on a regular basis, and several serious incidents and accidents in 1926 and 1927 highlighted the need for practical solutions to the icing problem [6]. Early research

[1]Supercooled water droplets are liquid water droplets that remain unfrozen despite being at temperatures below the freezing point.



(a) Aircraft engine [10] (b) Pitot probe [11]

Figure 1.1 Examples of ice accretion on aircraft components.

efforts were therefore mainly directed toward ice-protection systems (IPS)[2], including heated aircraft components, waterproof coatings, greased propellers, electrically heated pitot-static tubes, and pneumatic de-icing boots [6]. Although some of these solutions proved useful in specific applications, they remained largely empirical and did not provide a complete understanding of the physical mechanisms governing ice accretion.

Over the following decades, repeated claims that aircraft icing had been effectively solved proved premature, as operational experience continued to reveal unresolved challenges and the need for dedicated experimental investigation under controlled conditions. This need motivated the development of controlled icing research capabilities, culminating in the construction of the Icing Research Tunnel (IRT) in Cleveland in the 1940s [6]. The IRT enabled systematic investigations of supercooled cloud conditions, droplet impingement, aerodynamic performance degradation, and the performance of anti-icing and de-icing systems. These experimental studies played a central role in improving the physical understanding of aircraft icing and later provided essential data for the development, validation, and certification of numerical icing prediction methods [12].

1.2 Ice Morphologies

Depending on the atmospheric and aerodynamic conditions, different ice morphologies may develop on aircraft surfaces. At low temperatures, rime ice forms when supercooled droplets freeze almost immediately upon impact. This rapid solidification produces a relatively smooth, streamlined, and opaque ice shape. The opaque appearance of rime ice is asso-

[2]Ice protection systems were divided into two categories : Anti-icing systems are designed to prevent ice from forming on protected surfaces, whereas de-icing systems allow ice to form temporarily before removing it.

ciated with air trapped within the ice during the rapid freezing process.

As the temperature increases toward the freezing point, glaze ice may form. In this regime, the impinging droplets do not freeze completely upon impact. Instead, part of the liquid water remains on the surface, forming a thin water film that can run back before freezing downstream. Glaze ice is therefore usually denser and more transparent than rime ice, and it often leads to more complex ice shapes, including horn-like structures, and, on swept configurations, three-dimensional features such as scallops or lobster tails (Fig. 1.2).

A third regime, mixed ice, occurs as a transition between rime and glaze conditions. In this case, glaze-like ice may form near the stagnation region, where the impingement rate is high and the water does not freeze completely on impact. Farther downstream, the remaining droplets or runback water may freeze more rapidly, producing rime-like deposits. Mixed ice therefore combines features of both rime and glaze ice.

Figure 1.2 Ice accretion on the leading edge of the CRM hybrid wing model tested in the NASA Icing Research Tunnel. Photos taken from [1]

1.3 Regulations & Certification

The diversity of ice morphologies and the severity of their aerodynamic effects make in-flight icing a critical consideration in aircraft certification. Regulatory authorities require aircraft to demonstrate safe operation within prescribed icing environments, which are defined through envelopes of atmospheric and flight conditions. A major international reference for the certification of transport-category airplanes is Title 14 of the Code of Federal Regulations, Part 25, adopted by the Federal Aviation Administration (FAA). Within this framework, Appendix C [13] has historically defined the reference atmospheric icing envelopes used for the design and certification of IPS. These envelopes, in use since 1964, describe continuous maximum and intermittent maximum icing conditions in terms of parameters such as liquid water content, temperature, altitude, cloud horizontal extent, and representative droplet diameter [14]. They were primarily established for conventional cloud icing conditions, where the droplet median volume diameter (MVD) is typically limited to the range of standard supercooled droplets (SSD), up to approximately 50 μm.

However, several in-service events demonstrated that aircraft could also encounter icing environments outside the original Appendix C envelopes. A major example is the crash of American Eagle Flight 4184, an ATR-72, near Roselawn, Indiana, on October 31, 1994 [15]. The accident investigation identified icing as a central factor in the loss of control and highlighted the safety implications of supercooled large droplet (SLD) conditions, including freezing drizzle and freezing rain. These environments may contain droplets significantly larger than those considered in Appendix C, with droplet diameters reaching several hundred micrometres, typically up to about 500 μm in freezing-drizzle conditions, and up to approximately 1000 μm in freezing-rain conditions.

This recognition led to additional regulatory developments, culminating in the introduction of Appendix O to 14 CFR Part 25 for SLD icing conditions [16]. Appendix O extends the certification envelope to include freezing drizzle and freezing rain, thereby addressing icing scenarios in which large droplets may splash, rebound, or break up before freezing. These mechanisms may lead to ice formation downstream of the protected region. Consequently, aircraft seeking certification for flight in icing conditions must demonstrate safe operation under these conditions.

The certification of IPS may rely on several complementary approaches, including flight testing, icing wind-tunnel testing, and numerical simulation. Flight tests provide the most representative assessment of aircraft behavior in realistic operating conditions and may be conducted either in natural icing environments or using artificial ice shapes. However, such tests are expensive, difficult to reproduce, and may involve safety risks, particularly when degraded aerodynamic performance or controllability must be evaluated.

Icing wind tunnels therefore became an essential complementary experimental tool for studying ice accretion and evaluating IPS performance under controlled conditions. Facilities such as the NASA IRT, the National Research Council (NRC) Low-Speed Icing Wind Tunnel in

Canada [17], the Rail Tec Arsenal (RTA) facility in Austria [18], and the Office National d'Études et de Recherches Aérospatiales (ONERA) icing facilities [19] in France have contributed extensively to the development of experimental databases for icing research and certification support. Compared with flight testing, wind-tunnel experiments offer better control of atmospheric parameters, and detailed access to local measurements. Nevertheless, they remain costly and face important scaling limitations, especially as aircraft components increase in size. Reproducing full-scale icing conditions often requires preserving several competing similarity parameters, including the relationships between droplet size, leading-edge radius, liquid water content, velocity, temperature, and accretion time. In practice, the model geometry or operating conditions must often be adjusted to fit within the facility constraints, and these adjustments can themselves require dedicated extensive analyses [20].

In addition, ice accretion experiments are subject to variability from one test to another, owing to uncertainties in cloud calibration, droplet distributions, thermal conditions, surface roughness, and measurement procedures. These limitations have motivated the development of numerical icing tools as a complementary approach to flight and wind-tunnel testing.

1.4 Ice Accretion Numerical Modeling

Ice accretion results from the interaction of several physical processes, including the aerodynamic flow around the aircraft, the transport and impingement of supercooled water droplets, the local thermodynamic balance at the surface, and the progressive modification of the geometry due to ice growth.

Although ice accretion is an unsteady phenomenon, most computational icing solvers rely on a quasi-steady approximation. This assumption is justified by the separation between the short characteristic time scales of the airflow and droplet motion and the much longer time scale associated with ice growth. In practice, the total exposure time is divided into one or several discrete accretion steps. Within each step, all the physical processes are simulated under steady-state conditions.

The simplest implementation consists of performing the entire accretion process in a single step. This approach is attractive because it limits the computational cost and avoids repeated geometry and mesh updates. However, it assumes that the flow, droplet trajectories, and surface thermodynamic solutions remain representative throughout the full exposure time, which can reduce the accuracy of the predicted ice shape. A more representative strategy consists of using a multilayer approach, in which the geometry is updated several times during the accretion process and the physical fields are recomputed on the evolving surface. This improves the coupling between ice growth and the surrounding airflow, but also increases the computational cost and introduces additional robustness issues. In particular, repeated geometry deformation or remeshing may progressively deteriorate the mesh quality, leading to highly skewed, tangled, or invalid elements that can interrupt the simulation.

1.5 Problem Statement

Two-dimensional icing simulations have played a central role in the development and validation of ice accretion models. They remain computationally efficient and are particularly useful for airfoil configurations, parametric studies, and comparisons with well-controlled experimental databases. In some applications, they are also extended to approximate three-dimensional effects through 2.5D approaches [21], for example by applying sectional corrections on swept wings. However, such methods remain limited when the ice accretion process is governed by genuinely three-dimensional flow features, spanwise transport, complex geometry effects, or non-uniform impingement patterns.

Fully three-dimensional simulations provide a more general framework for predicting ice accretion on realistic aircraft components and complex configurations. Nevertheless, they are significantly more demanding in terms of computational resources and numerical robustness. These difficulties become particularly important in multilayer simulations, where the geometry must be updated repeatedly as the ice grows. Each update may degrade the mesh quality or generate invalid surface and volume elements. As a result, a simulation that has already required substantial computational effort may fail before reaching the final accretion time.

This robustness constraint partly explains why single-step ice accretion strategies remain widely used in practical applications. They are simpler to execute and reduce the risk of mesh failure, but they also limit the ability of the solver to account for the feedback between the evolving ice shape and the surrounding flow and droplet fields. In the Ice Prediction Workshop (IPW)[3], many submitted ice-shape predictions relied on such single-step approaches. While effective from a computational standpoint, these methods generally showed reduced ability to reproduce complex accretion features, especially when compared with multilayer simulations. This highlights the need for numerical strategies capable of preserving the robustness of single-step approaches while retaining the improved physical fidelity of multilayer ice-growth modeling.

[3]The Ice Prediction Workshops (IPW) [22, 23] is an AIAA international benchmark initiative for evaluating numerical aircraft icing tools against experimental data.

1.5.1 Motivation for the Present Work

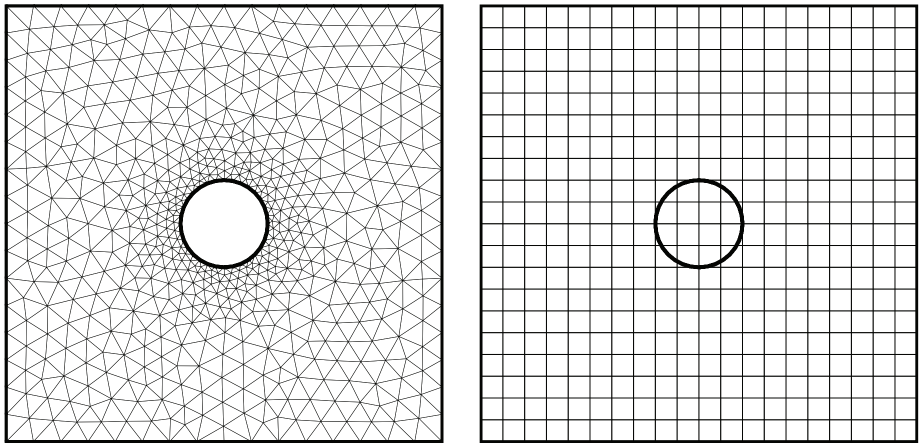
The preceding discussion identifies the repeated update of the iced geometry as a major limitation of three-dimensional multilayer ice accretion simulations. Although the flow, droplet, and thermodynamic solutions must be recomputed consistently as the ice shape evolves, the ability to reach the final accretion time often depends on the robustness of the geometry-update procedure itself.

This motivates the use of geometry representations that are less sensitive to local surface irregularities. Previous studies [2,24,25] have demonstrated the potential of level-set descriptions for representing and evolving iced geometries in two-dimensional configurations. In a level-set formulation, the interface is not tracked explicitly by the surface mesh. Instead, it is represented implicitly as the zero level of a scalar function ϕ. The sign of ϕ distinguishes the ice and air regions, while the interface is recovered from the condition $\phi = 0$.

Building on these first attempts, the present work extends this idea to three-dimensional multilayer ice accretion simulations, where geometry updates become more challenging due to spanwise variations and increasingly complex ice features. By representing the iced surface implicitly, the geometry can be displaced and reconstructed in a more systematic manner, while preserving the clean portions of the geometry that are not modified by ice accretion. This reconstruction strategy reduces the sensitivity of the workflow to local geometric defects and provides greater control over the quality of the reconstructed surface mesh.

Since geometry evolution is only one component of the icing simulation chain, the preceding physical models must also provide reliable inputs to the accretion process. In particular, droplet impingement predictions determine the local water mass flux supplied to the thermodynamic model and directly influence the ice-growth rate. For this reason, Eulerian and Lagrangian droplet formulations, including their extension to Supercooled Large Droplet conditions, are also examined in this thesis.

Finally, immersed-boundary methods are considered in order to reduce the dependence on body-fitted volume remeshing. In a body-fitted approach, the computational mesh conforms to the solid surface, as shown in Figure 1.3a. In contrast, an immersed-boundary method embeds the geometry within a non-conformal background mesh, as shown in Figure 1.3b. The geometry is therefore represented independently of the mesh, which can simplify mesh generation and geometry handling for complex or evolving iced surfaces. The approach is assessed by comparing its aerodynamic, droplet-impingement, and ice-accretion predictions with those obtained using established body-fitted formulations.



(a) Body-fitted mesh. (b) Immersed-boundary mesh.

Figure 1.3 Mesh representation around a cylinder using (a) a body-fitted mesh and (b) an immersed-boundary mesh. In the body-fitted approach, the mesh conforms to the cylinder surface, whereas in the immersed-boundary approach, the cylinder is embedded within a non-conformal background mesh.

1.6 Objectives of the Present Study

The primary objective of this thesis is to improve the robustness of three-dimensional multilayer ice accretion simulations through level-set-based geometry evolution. The proposed developments aim to reduce geometric failures during the accretion process, improve the quality of reconstructed iced surfaces, and maintain reliable aerodynamic, droplet-impingement, thermodynamic, and ice-shape predictions throughout the icing simulation chain.

More specifically, this work develops and assesses numerical methods required to support robust multilayer ice accretion simulations, including Lagrangian particle tracking for droplet impingement prediction, Supercooled Large Droplet effects in Eulerian and Lagrangian formulations, level-set-based geometry update and surface reconstruction, and an immersed-boundary extension of the existing icing framework.

To accomplish this objective, the following sub-objectives are defined.

Droplet Impingement Modeling and Supercooled Large Droplet Effects

O.1 Develop, accelerate, and assess Lagrangian particle-tracking methods for droplet impingement prediction

The first objective is to develop and evaluate Lagrangian particle-tracking methods for predicting droplet trajectories and collection efficiency. The method is assessed in terms of impingement limits, collection-efficiency distributions, and computational cost, and is compared with the existing Eulerian droplet model.

O.2 Incorporate Supercooled Large Droplet effects in Eulerian and Lagrangian formulations

The second objective is to account for Supercooled Large Droplet effects in both Eulerian and Lagrangian impingement models. These developments are used to assess the influence of large-droplet effects on droplet collection, impingement limits, and ice accretion.

Multilayer Ice-Shape Evolution

O.3 Improve the robustness of three-dimensional multilayer ice accretion simulations using a level-set formulation

The third objective is to improve the robustness of multilayer ice-shape evolution using a level-set description of the evolving geometry. The method is intended to support geometry displacement, surface reconstruction, and preservation of the clean portions of the geometry, while reducing geometric failures during multilayer simulations.

O.4 Evaluate the level-set-based methodology in representative icing configurations

The fourth objective is to apply the level-set-based multilayer approach to representative two-dimensional and three-dimensional icing configurations. These cases are used to evaluate the accuracy, robustness, computational efficiency, and reconstructed surface mesh quality of the proposed geometry-evolution strategy.

Immersed-Boundary Formulation and Solver Integration

O.5 Explore immersed-boundary formulations for ice accretion simulations on evolving geometries

The fifth objective is to extend an existing immersed-boundary flow solver by integrating droplet impingement, thermodynamic ice accretion, and geometry-evolution capabilities within a unified ice-accretion framework. This extension is intended to reduce the dependence on repeated body-fitted volume remeshing while providing a robust platform for simulations involving complex and evolving iced geometries.

O.6 Assess the immersed-boundary approach against body-fitted formulations

The sixth objective is to assess the ability of the immersed-boundary approach to reproduce body-fitted predictions for selected verification and validation cases. The comparison focuses on droplet collection efficiencies, and final ice shapes in order to identify the accuracy and limitations of the proposed formulation.

1.7 Thesis Outline

This thesis is organized as a thesis by articles. Chapter 2 first presents the state of the art in numerical ice accretion modeling. It reviews the main components of icing simulation workflows, including aerodynamic modeling, droplet impingement, thermodynamic ice-growth models, SLD effects, geometry evolution strategies, immersed-boundary methods, and level-set techniques.

The first article focuses on Lagrangian particle tracking for droplet impingement prediction under Supercooled Large Droplet conditions. Particular attention is given to the treatment of large-droplet effects and to the ability of the method to accurately and efficiently predict collection-efficiency distributions.

The second article addresses the problem of robust geometry evolution in three-dimensional multilayer ice accretion simulations. A level-set-based approach is developed and assessed as a means of updating complex ice shapes while reducing the risk of mesh degradation and invalid geometries during the accretion process.

The third article combines the developments of the previous chapters within an immersed-boundary framework for ice accretion simulations. In this formulation, the aerodynamic and droplet calculations are performed on a background mesh, while the evolving iced geometry is handled through level-set-based reconstruction and update procedures. The objective is to evaluate whether such a framework can reduce the meshing constraints traditionally associated with body-fitted multilayer icing simulations.

Finally, the thesis concludes with a synthesis of the main contributions, followed by a discussion of the limitations of the proposed methods and recommendations for future work.

CHAPTER 2 LITERATURE REVIEW

This chapter reviews the principal numerical ingredients involved in aircraft ice-accretion simulations. It first presents the conventional quasi-steady framework adopted by most icing solvers before reviewing the aerodynamic, dispersed-phase, thermodynamic, geometry-evolution, and geometry-representation models and techniques reported in the literature. The chapter concludes with a synthesis of the current state of the art and identifies the remaining challenges that motivate the developments presented in the following chapters.

2.1 Overview of Numerical Ice Accretion Modeling

Aircraft icing is the accumulation of ice on aircraft surfaces caused by the impingement and freezing of supercooled water droplets. From a numerical point of view, the problem involves the coupling of an external aerodynamic flow, a dispersed liquid phase, a surface thermodynamic balance, and an evolving fluid-solid interface. These different components are characterized by distinct physical time scales, which motivates the modular formulation used in most ice-accretion solvers [26].

Certification icing conditions generally involve exposure over several minutes, resulting in an ice-accretion time scale that is several orders of magnitude longer than the characteristic flow and droplet time scales [26]. Owing to this disparity, the transient ice-accretion process may be approximated as a succession of quasi-steady states.

In addition to this separation of time scales, the low concentration of liquid water in icing clouds permits a further simplification in the treatment of the dispersed phase. Certification icing envelopes involve supercooled droplets with characteristic diameters ranging from approximately 10 to 50 μm, while SLD conditions can extend to diameters approaching 1000 μm. For a typical liquid water content (LWC) of 1 g/m^3, the corresponding liquid volume fraction is of the order of 10^{-6} [27, 28]. This small volume fraction justifies the common assumption of one-way coupling between the air and droplet phases.

These assumptions provide the foundation for the conventional quasi-steady ice-accretion framework (Fig. 2.1). In its simplest form, the aerodynamic flow is first computed around the clean geometry. The resulting velocity field is then used to determine the transport of the dispersed phase and the local droplet impingement rate on the exposed surface. The latter is then provided to a surface thermodynamic model, which evaluates the local mass and energy balances and returns the ice-growth rate. Finally, the geometry is updated

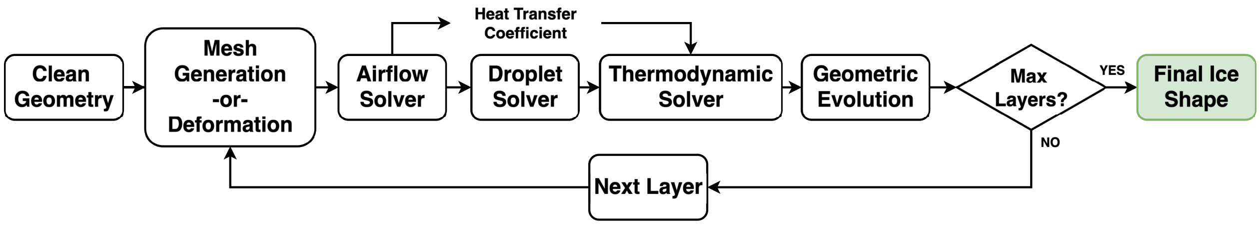


Figure 2.1 Schematic of the multilayer ice accretion framework.

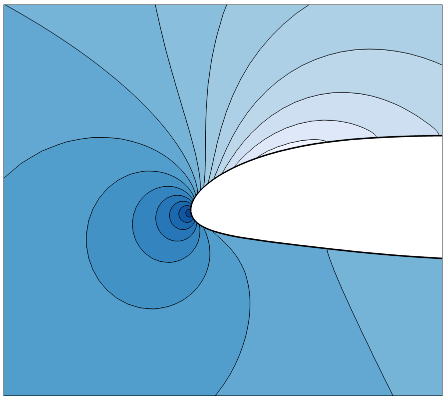
according to the predicted ice thickness. Representative aerodynamic, droplet, geometry-evolution, and mesh-update solutions associated with this sequence are shown in Fig. 2.2. When this sequence is performed only once over the full exposure time, the approach is often referred to as a single-step or single-shot accretion simulation. When the total icing time is divided into several shorter intervals, the same sequence is repeated on the evolving geometry, leading to a multilayer formulation. Intermediate strategies, such as predictor–corrector formulations [29, 30], have also been proposed to partially account for the retroaction of the growing ice shape on the flow and droplet fields. This limits the additional computational cost to a small number of surface-correction cycles.

Although the quasi-steady multilayer approach remains the standard strategy in engineering ice-accretion simulations, the physical process is inherently unsteady. Fully time-accurate formulations have been reported in the literature [31]. More recently, unsteady couplings involving scale-resolving flow simulations, such as LES, have also been explored [32]. These approaches provide a more direct representation of the continuous interaction between the flow and the evolving ice surface, but their computational cost remains prohibitive for routine industrial icing applications, especially in three-dimensional configurations.

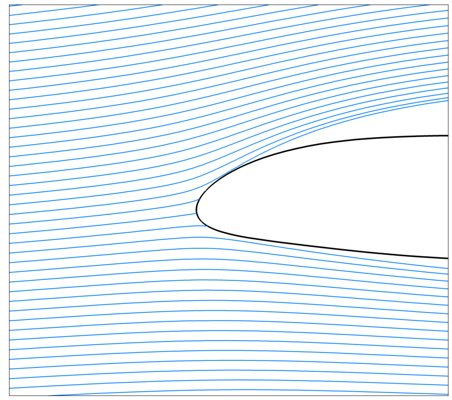
To provide a broader perspective on existing numerical approaches, Table 2.1 summarizes several state-of-the-art ice-accretion solvers reported in the literature [22, 23]. The table highlights the main numerical ingredients used in each framework. The remainder of this chapter then examines these components in more detail.

Table 2.1 Comparison of representative numerical aircraft icing solvers.

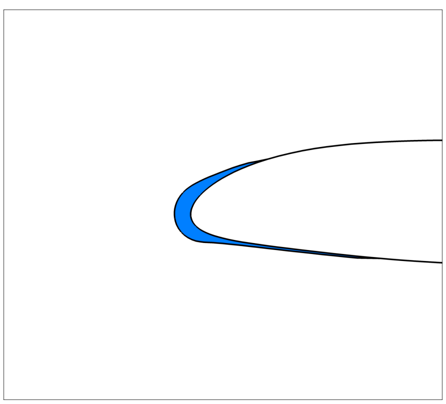
Organization	Solver	Flow	Droplet	Thermo	Geo Evolution	Geo Rep.	Multishot
NASA	LEWICE	Panel / Euler / RANS	Lagrangian	Messinger / Modified Messinger	Lagrangian	Body-fitted	✓
NASA	GlennICE	RANS	Lagrangian	Messinger	Face Offset	Body-fitted	✓
ANSYS Canada	FENSAP-ICE	RANS	Eulerian	Modified Messinger	Lagrangian	Body-fitted	✓
ONERA	ONERA IGLOO3D	RANS	Eulerian	Extended Messinger	Lagrangian	Body-fitted	✓
Bombardier	Dragon Ice Suite	RANS	Eulerian	Iterative Messinger	Hyperbolic Marching	Body-fitted	✓
Polytechnique Montréal	CHAMPS-ICE	RANS	Eulerian	Iterative Messinger	Lagrangian / RBF	Body-fitted	✓
CIRA	SIMBA	IBM	Eulerian	Iterative Messinger	Lagrangian	Immersed Boundary	✓
TU Braunschweig	DICEPS	RANS	Eulerian	–	Lagrangian	Body-fitted	✓
University of Twente	SWIM	RANS	Eulerian	SWIM	Lagrangian	Body-fitted	×
Politecnico di Milano	PoliMIce	RANS	Lagrangian	Modified Myers	Level Set	Body-fitted	✓



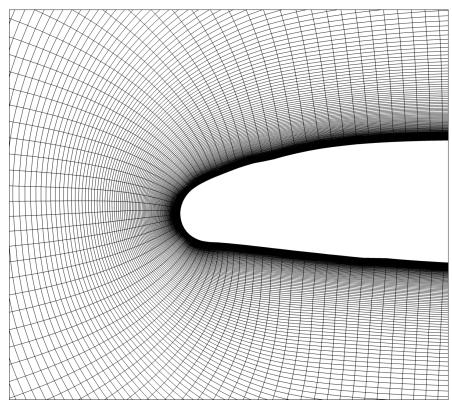
(a) Aerodynamic solution represented by pressure contours.



(b) Droplet solution represented by particle trajectories.



(c) Geometry evolution represented by the accreted ice shape.



(d) Updated computational mesh following geometry evolution.

Figure 2.2 Illustration of the principal stages involved in a aircraft ice accretion simulation: (a) aerodynamic flow solution, (b) droplet impingement prediction, (c) geometry evolution through ice accretion, and (d) mesh update for the subsequent accretion layer.

2.2 Aerodynamics

Early theoretical developments in aircraft icing were established before the availability of modern computational resources. Foundational contributions by Langmuir and Blodgett, Hardy, and Messinger provided the theoritical basis for ice accretion modeling [33]. However, these early models were restricted to simple configurations, such as cylinders and spheres, or relied on empirical corrections when analytical solutions were not available. The extension of icing analysis to more representative aerodynamic geometries became possible only with the development of computational methods.

The first generation of numerical icing tools emerged from research efforts led mainly by NASA Lewis Research Center in the United States, the Royal Aerospace Establishment in the United Kingdom, and ONERA in France [33]. Early codes such as LEWICE [34] and CANICE [35] relied primarily on inviscid aerodynamic descriptions, typically based on potential-flow or panel methods, coupled with boundary-layer corrections to estimate the near-wall quantities required by the icing model. Similarly, ONERA developed icing tools based on Euler flow solutions combined with boundary-layer calculations [36]. These approaches provided an efficient framework for ice-accretion prediction and formed the basis of several widely used engineering tools.

With the maturation of computational fluid dynamics (CFD), especially from the 1970s onward, Reynolds-averaged Navier–Stokes (RANS) solvers progressively became the dominant aerodynamic model for practical icing simulations [37]. RANS solvers provide a more complete description of viscous effects and near-wall flow features, which are important for the ice shape prediction. This transition is reflected in recent IPW, where the majority of reported simulations rely on RANS-based aerodynamic solutions, most often using turbulence closures such as the Spalart–Allmaras or k–ω models [38, 39]. Although RANS calculations are more expensive than inviscid approaches, they remain affordable for industrial applications and provide a practical compromise between physical fidelity and computational cost.

At the same time, recent work has renewed interest in lower-cost inviscid formulations strongly coupled with boundary-layer models [25, 40, 41]. These approaches aim to recover part of the accuracy of RANS-based icing simulations [42, 43] while retaining the computational efficiency of Euler or potential-flow solvers. Such strategies are particularly attractive for design studies, parametric analyses, and multilayer ice-accretion calculations. The current literature therefore reflects two complementary trends: the continued use of RANS solvers as the standard aerodynamic model for engineering icing simulations, and the renewed interest in revisiting earlier inviscid and low- to medium-fidelity approaches using the knowledge and reference data gained from high-fidelity computational tools.

2.3 Dispersed Phase Modeling

Droplet-impingement prediction provides the local impingement rate required by the thermodynamic stage. Before the local ice-growth rate can be evaluated, the amount of supercooled water reaching the surface must first be determined. This quantity is commonly expressed through the collection efficiency, denoted by β, a dimensionless variable that represents the ratio of the local impinging water mass flux to the freestream water mass flux. The dimensional impinging water mass flux can therefore be written as

$$\dot{m}_{\mathrm{imp}} = \beta \, \mathrm{LWC} \, U_\infty \tag{2.1}$$

where U_∞ is the freestream velocity. The collection efficiency directly determines the water mass available for freezing or runback and therefore strongly influences the predicted ice shape.

The definition of β depends on the droplet formulation. In an Eulerian framework, the droplet phase is treated as a continuous medium. The local collection efficiency is then obtained from the normal component of the droplet mass flux at the wall [44]. In nondimensional form, it can be expressed as

$$\beta = -\alpha \, \boldsymbol{u}_d \cdot \boldsymbol{n} \tag{2.2}$$

where α is the nondimensional water concentration, $\boldsymbol{u}_d$ is the droplet velocity, and $\boldsymbol{n}$ is the outward unit normal to the surface. In a Lagrangian framework, the collection efficiency is instead determined by tracking the deformation of a droplet streamtube initialized in the freestream as it approaches and intersects the surface. As illustrated in Fig. 2.3, β may be interpreted, for a given surface element, as the ratio between the upstream seeding area associated with a set of neighboring droplet trajectories and the corresponding impingement area on the wall [45]:

$$\beta = \frac{A_{\mathrm{upstream}}}{A_{\mathrm{impingement}}} \tag{2.3}$$

Historically, Lagrangian droplet tracking was the only approach in early icing codes [34, 35]. In solvers based on potential-flow or panel methods, the airflow field was inexpensive to evaluate and droplet trajectories could be integrated directly through the computed velocity

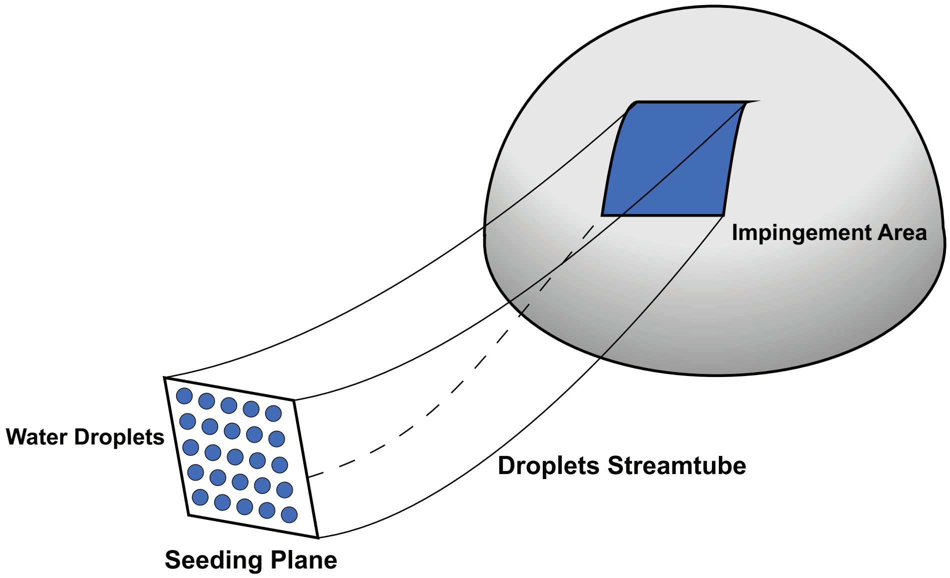


Figure 2.3 Schematic representation of the Lagrangian droplet-seeding procedure. Droplets are released from an upstream plane and tracked until they either impinge on the immersed surface or bypass the body.

field. This made the Lagrangian formulation attractive because it provided a clear physical description of droplet motion. However, as ice-accretion solvers evolved toward Euler and RANS aerodynamic models, the computational cost of Lagrangian tracking became more restrictive. Accurate impingement maps require a large number of droplets, particularly in three-dimensional configurations. In addition, each droplet trajectory requires repeated interpolation of the carrier-flow solution, localization of the droplet within the computational mesh, and detection of surface impacts. These operations can become expensive and difficult to parallelize efficiently on complex unstructured grids. These limitations motivated the development of Eulerian droplet formulations [44], in which the dispersed phase is solved as a continuum using partial differential equations (PDE).

2.3.1 Eulerian Droplet Formulation

Eulerian dispersed-phase models describe the droplet field through continuous variables defined over the computational domain [44]. The main unknowns are the nondimensional water concentration α and the droplet velocity $\boldsymbol{u}_d$. A common formulation consists of a conservation equation for the droplet concentration

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \boldsymbol{u}_d) = 0 \tag{2.4}$$

combined with a momentum equation for the droplet phase

$$\frac{\partial(\alpha\boldsymbol{u}_d)}{\partial t} + \nabla\cdot(\alpha\boldsymbol{u}_d\otimes\boldsymbol{u}_d) = \frac{C_D Re_D}{24K}\alpha(\boldsymbol{u}_a-\boldsymbol{u}_d) + \alpha\left(1-\frac{\rho_a}{\rho_w}\right)\frac{\boldsymbol{g}}{Fr^2} \tag{2.5}$$

where $\boldsymbol{u}_a$ is the airflow velocity, C_D is the droplet drag coefficient, Re_D is the droplet Reynolds number (Eq. 2.6), K is the inertia parameter (Eq. 2.7), ρ_a and ρ_w are the air and water densities, respectively, $\boldsymbol{g}$ is the gravitational acceleration, and Fr is the Froude number (Eq. 2.8).

$$\mathrm{Re}_d = \frac{\rho_a\left\|\boldsymbol{u}_a-\boldsymbol{u}_d\right\|D}{\mu_a} \tag{2.6}$$

$$K = \frac{\rho_w D^2 U_\infty}{18\mu_a L} \tag{2.7}$$

$$\mathrm{Fr} = \frac{U_\infty}{\sqrt{Lg}} \tag{2.8}$$

The Eulerian approach is well suited to modern finite-volume CFD solvers. Since the droplet equations are written in conservative form, they can be discretized using numerical techniques similar to those used for the airflow equations. This makes the approach relatively efficient, naturally compatible with parallel computing, and convenient for three-dimensional simulations on complex grids.

Nevertheless, the Eulerian formulation has limitations. Since the droplet phase is averaged within each computational cell, individual droplet trajectories, impact velocities, and impact angles are not directly available. Moreover, numerical diffusion may smear the impingement limits, especially near shadow zones. Numerical oscillations may also appear in low-concentration regions and can lead to nonphysical values of α. Additional numerical treatments are therefore often required to preserve positivity, reduce artificial diffusion, and improve conservation properties [46].

2.3.2 Lagrangian Droplet Formulation

In a Lagrangian formulation, droplets are tracked individually by integrating their equations of motion through the carrier-flow field. The droplet velocity $\boldsymbol{u}_d$ and position $\boldsymbol{x}_d$ satisfy

$$\frac{d\boldsymbol{u_d}}{dt} = \frac{\mathbf{f}}{m} + \mathbf{g} = F(\boldsymbol{u_a}-\boldsymbol{u_d}) + \frac{\rho_d-\rho_a}{\rho_d}\mathbf{g} \tag{2.9}$$

and

$$\frac{d\boldsymbol{x}_d}{dt} = \boldsymbol{u}_d \tag{2.10}$$

where F is a scalar drag coefficient :

$$F = \frac{3C_D\rho_a\left|\boldsymbol{u}_a - \boldsymbol{u}_d\right|}{4\rho_w D} \tag{2.11}$$

Although Lagrangian methods are considered expensive for high-fidelity icing simulations, they have recently gained renewed interest [47–51]. Modern high-performance computing resources, improved particle-localization algorithms, and the need for detailed impact information have made Lagrangian approaches increasingly attractive. This is particularly true for SLD conditions, where splashing, bouncing, and secondary droplet motion are more naturally described in the Lagrangian framework.

2.3.3 Supercooled Large Droplet Modeling

Supercooled Large Droplet conditions introduce additional physical mechanisms into droplet-impingement modeling. Compared with SSD, SLDs have larger diameters and therefore greater inertia, making their trajectories less responsive to changes in the surrounding airflow [26]. They may consequently impinge farther downstream and potentially outside regions protected by conventional IPS.

The additional phenomena associated with SLDs may be separated into effects occurring before and during impact. In-flight deformation and breakup can be regarded as second-order effects because they indirectly influence impingement by modifying the droplet shape, aerodynamic drag, and trajectory before contact with the surface. By contrast, splashing, bouncing, and secondary-droplet ejection constitute first-order wall-interaction effects because they directly determine the fraction of the impinging water mass that remains on the surface [52].

As an SLD travels through a nonuniform flow field, variations in the aerodynamic pressure and shear stresses acting over its surface may deform it from its initially spherical shape. The resistance of the droplet to this deformation is governed primarily by surface-tension and viscous forces. When the aerodynamic forces become sufficiently large relative to these restoring forces, the spherical-droplet assumption is no longer valid, and the droplet may adopt an oblate or more complex nonspherical shape. Under more severe conditions, continued deformation may ultimately lead to breakup into smaller droplets [53].

Droplet deformation can affect impingement by modifying the aerodynamic drag and, consequently, the droplet response time and trajectory. Breakup may additionally alter the droplet-size distribution and produce smaller fragments that interact more strongly with the airflow. Nevertheless, the influence of deformation and breakup on the resulting impingement distribution has generally been found to remain limited for many icing conditions, with potentially greater importance under freezing-drizzle conditions [52, 54]. For this reason, these mechanisms are often neglected or represented through simplified corrections rather than modeled explicitly.

A commonly used approximation accounts for deformation by modifying the drag coefficient without explicitly resolving the evolving droplet shape. The effective drag coefficient may be obtained by blending correlations for spherical droplets and oblate disks as a function of the droplet Reynolds and Weber numbers [55]:

$$C_D = (1 - f)\, C_{D,\text{sphere}} + f C_{D,\text{disk}} \tag{2.12}$$

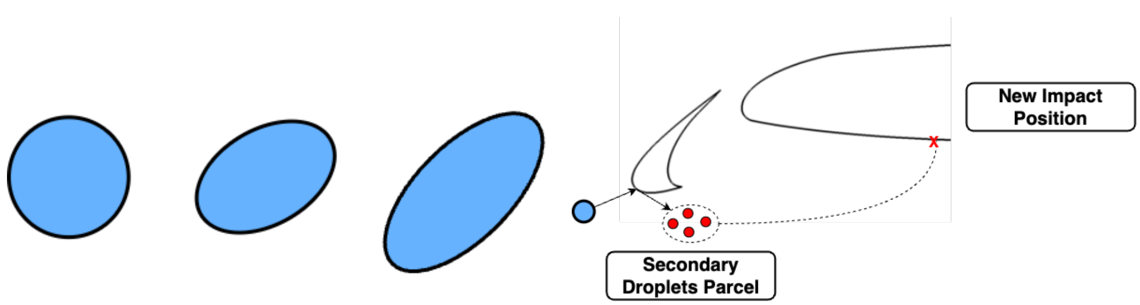
with

$$f = 1 - \left(1 + 0.07\sqrt{\text{We}}\right)^{-6} \tag{2.13}$$

where the Weber number is defined as

$$\text{We} = \frac{\rho_a d U_{\text{rel}}^2}{\sigma} \tag{2.14}$$

where ρ_a is the air density, d is the droplet diameter, U_{rel} is the relative air–droplet velocity, and σ is the surface tension of water. $C_{D,\text{sphere}}$ and $C_{D,\text{disk}}$ depend on the droplet Reynolds number.



(a) Droplet Deformation. (b) Droplet splashing.

Figure 2.4 Droplet Phenomena.

The most consequential SLD effects generally arise during droplet-wall interaction. At relatively low impact energy, an impinging droplet may stick to or spread over the surface and contribute directly to the surface water film. Under other conditions, it may bounce, splash, or break into secondary droplets [56, 57]. These secondary droplets can be ejected from the surface and may subsequently reimpinge farther downstream, thereby extending the effective impingement region. The amount of water deposited on the surface is therefore determined not only by the primary collection efficiency, but also by the outcome of the impact process.

Because resolving individual droplet impacts with interface-capturing methods would be prohibitively expensive at aircraft scale [58], these first-order SLD wall-interaction effects are generally represented through semi-empirical models. Such models are commonly formulated in terms of nondimensional impact parameters, including the Reynolds, Weber, and Ohnesorge numbers, and may also account for the impact angle and impact velocity. A common approach consists in correcting the primary collection efficiency using a sticking rate ϵ, or equivalently a mass-loss coefficient f_m:

$$\beta_{\text{final}} = \epsilon\beta_0 = (1 - f_m)\,\beta_0, \tag{2.15}$$

where β_0 is the collection efficiency obtained before accounting for splashing or bouncing. This correction represents the fraction of the primary impinging water mass that remains on the surface after impact.

More advanced SLD models also account for the generation and reimpingement of secondary droplets. In a Lagrangian framework, this can be handled by reinjecting secondary droplets with prescribed diameters, velocities, and ejection directions obtained from semi-empirical correlations [50]. In an Eulerian framework, secondary reimpingement is more difficult to represent directly because the dispersed phase is described as a continuous field. One strategy consists in modifying the boundary conditions on splashing surfaces so that the secondary droplets are reemitted into the computational domain and transported by the Eulerian equations [59, 60]. The resulting collection efficiency may then include both the retained primary contribution and the mass associated with secondary reimpingement:

$$\beta_{\text{final}} = \epsilon\beta_0 + \sum \beta_{\text{reimp}}, \tag{2.16}$$

where β_{reimp} denotes the collection-efficiency contribution associated with reimpinging secondary droplets. The modeling of SLD effects therefore remains closely linked to the selected droplet formulation. Eulerian methods provide an efficient and robust framework for primary impingement calculations, but require additional boundary-condition treatments to represent secondary droplets. Lagrangian methods are generally more expensive, but naturally provide the impact quantities required by splashing and bouncing correlations and allow secondary droplets to be tracked explicitly.

Several semi-empirical SLD impact models have been proposed in the literature. Mundo et al. [61] developed an early splashing model describing the size and number of secondary droplets. Trujillo et al. [62] introduced a formulation based on impact parameters and surface-roughness effects. Honsek et al. [63] subsequently adapted this approach for implementation in FENSAP-ICE. Wright et al. [52] introduced an impact-angle dependence into the splashing criterion and calibrated the model using experimental water-collection data. Trontin et al. [54] proposed an alternative formulation based on separate functions describing the dependence of mass loss on impact angle and kinetic energy. Among these approaches, the models of Wright et al. [52] and Trontin et al. [54] remain the most widely adopted in aircraft-icing codes. A summary of representative splashing models is provided in Table 4.2.

Table 2.2 SLD splashing models.

Source	**Mass Loss Coefficient**	**Splashing Criterion**
Mundo et al. [61]	$n_s \left(\dfrac{d_s}{d_0}\right)^3$	$K_M = \left(Oh^{-0.4} We\right)^{0.625} > 57.7$
Trujillo et al. [62]	$0.8\left[1 - \exp\left(-0.85(K_y - 17)\right)\right]$	$K_y = \dfrac{1}{0.45^{3/8}}\left(Oh^{-0.4} We\right)^{0.3125} > 17$
Honsek et al. [63]	$\dfrac{3.8}{K_y^{1/2}}\left[1 - \exp\left(-0.85(K_y - 17)\right)\right]$	$K_y > 17$
Wright et al. [52]	$0.7(1 - \sin\theta_0)\left[1 - \exp\left(-0.0092(K_M - 200)\right)\right]$	$\dfrac{0.859\, K_M^{1/2}\left(\dfrac{\rho_w}{LWC}\right)^{1/8}}{\sin^{5/4}\theta_0} > 200, \qquad \theta < \dfrac{\pi}{2}$
Trontin et al. [54]	$1 - g\left(\dfrac{\theta}{\theta_c}\right) f\left(\dfrac{K_c - K_0}{K_0}\right)$	$K_c = We_n\, Oh^{-2/5} > 657$

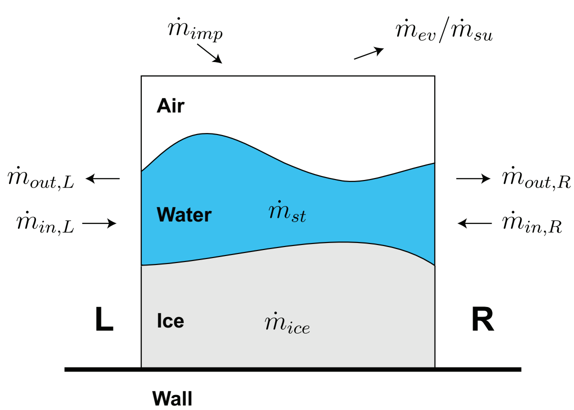
2.4 Thermodynamic Modeling

Once the airflow and droplet-impingement solutions have been obtained, a thermodynamic analysis is performed over the exposed surface to determine the local ice-accretion rate. The aerodynamic solution provides the convective heat-transfer coefficient, the wall shear stress, and the surface-pressure distribution, whereas the dispersed-phase solution provides the local impinging-water mass flux and the kinetic energy of the droplets at impact. These quantities are subsequently introduced into surface mass- and energy-balance equations describing the freezing, evaporation, and transport of water over the surface.

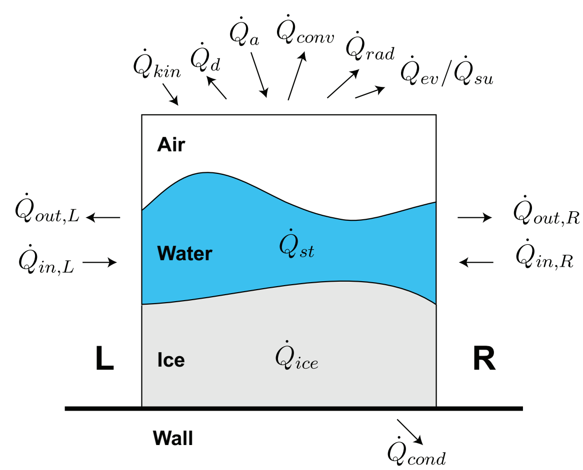
The thermodynamic problem involves three unknowns: the local ice-accretion rate, $\dot{m}_{\text{ice}}$; the outgoing runback-water flow rate, $\dot{m}_{\text{out}}$; and the surface temperature, T_s. The mass and energy conservation equations alone are insufficient to determine these quantities simultaneously. The system is therefore closed by introducing phase-state conditions derived from the classical Messinger model [64]:

- **Dry regime:** a dry element containing no liquid water;
- **Rime or fully freezing regime:** an element for which all incoming water freezes and $\dot{m}_{\text{out}} = 0$;
- **Glaze or partially freezing regime:** an element covered by a liquid-water film, for which only part of the incoming water freezes and $T_s = T_{\text{freeze}}$;
- **Fully liquid regime:** an element for which no ice is formed and $\dot{m}_{\text{ice}} = 0$.

Although the model remains the basis of many contemporary ice-accretion solvers, its original implementation relied on a surface-marching procedure initiated near the stagnation point. This dependence on a clearly identified stagnation location complicates its application to configurations involving multiple stagnation regions, separated flows, or general three-dimensional geometries.



(a) Mass balance.



(b) Energy balance.

Figure 2.5 Control volume diagrams for mass and energy balance equations. Adapted from [2].

Several extensions have consequently been proposed to improve either the physical description or the numerical robustness of the thermodynamic calculation. Myers et al. [65], for example, extended the classical formulation by accounting for heat conduction through the growing ice layer. Although this modification provides a more detailed representation of the temperature distribution within the ice, it does not remove the dependence of the solution on the treatment of the stagnation point.

A different strategy was introduced by Zhu et al. [66] through an iterative implementation of the Messinger model. Rather than relying exclusively on a predefined marching direction originating from a stagnation point, the surface-water distribution and thermodynamic state are updated iteratively. The transport direction of the runback water is determined from the local wall-shear-stress direction, with the surface-pressure gradient providing an additional contribution. This formulation is better suited to complex and three-dimensional geometries and has consequently been adopted by several modern ice-prediction tools, as reflected in the methods reported during recent ice-prediction workshops [22, 23].

Bourgault et al. [67] proposed an alternative formulation known as the Shallow-Water Icing Model (SWIM). In this approach, the liquid-water film is represented through surface PDEs describing its transport and evolution. In contrast to the iterative Messinger formulation, SWIM retains an explicitly unsteady description of the water film. It can therefore represent transient film-transport phenomena more naturally, but generally requires a more involved numerical treatment and introduces an additional computational cost. For many conventional ice-accretion conditions, the resulting ice shapes have nevertheless been found to remain broadly comparable to those obtained using iterative Messinger formulations [26]. A more comprehensive assessment and comparison of the classical Messinger model, its extensions, and SWIM formulations is provided by Lavoie et al. [68].

2.5 Geometry Evolution

Once the local ice-accretion rate has been determined by the thermodynamic solver, the solid surface must be updated to represent the newly accumulated ice. The updated surface then becomes the aerodynamic geometry for the subsequent accretion layer. Geometry-evolution techniques can generally be divided into two broad categories: interface-tracking methods, in which the ice–air interface is represented explicitly, and interface-capturing methods, in which the interface is embedded implicitly within a field defined over a computational domain.

Interface Tracking

Interface-tracking methods represent the ice–air interface using marker points located directly on the surface or within the surrounding control volumes. In ice-accretion simulations, the markers generally correspond to the nodes of the surface mesh. The position of the interface is therefore known explicitly throughout the geometry-update procedure. The most commonly used approach is the algebraic geometry-evolution method, also referred to as Lagrangian node displacement [69]. In its simplest form, each surface node is moved along the local normal by the predicted ice thickness:

$$\boldsymbol{x}_{\text{new}} = \boldsymbol{x}_{\text{old}} + h_{\text{ice}}\boldsymbol{n} \tag{2.17}$$

where $\boldsymbol{x}_{\text{old}}$ and $\boldsymbol{x}_{\text{new}}$ denote the initial and updated node positions, respectively, h_{ice} is the local accumulated ice thickness, and $\boldsymbol{n}$ is the outward surface-normal vector. Several extensions have been introduced to improve the robustness and accuracy of this displacement procedure. One approach consists of dividing the prescribed ice thickness into a sequence of smaller increments and recomputing the surface normals after each intermediate displacement [70]. This reduces the error associated with extrapolating the initial normals over relatively large accretion distances. Another formulation that has been proposed consists of deriving the displacement direction from a normal-vector field rather than directly from the discrete surface normals [29, 40]. Such fields are typically obtained from the gradient of a distance function satisfying an Eikonal equation [71]. Since the distance field is computed over the surrounding domain, neighboring displacement directions become naturally aligned and vary smoothly across the surface. This reduces the sensitivity of the geometry update to local irregularities in the surface triangulation, particularly in regions of high curvature.

Bourgault et al. [2] proposed a different interface-tracking approach based on constrained hyperbolic mesh generation. In this method, successive mesh layers are generated outward from the initial surface until the prescribed ice thickness is reached. An advantage of this formulation is that the newly accreted region is directly represented by a volume mesh, allowing the added ice volume to be evaluated without requiring a separate reconstruction procedure. Its application is nevertheless restricted to structured surface meshes, which limits its generality for complex three-dimensional configurations.

The principal advantage of interface-tracking methods is that the interface position is explicitly available, allowing fine geometric structures to be represented accurately when the surface discretization is sufficiently resolved. However, large node displacements may lead to surface-element distortion, self-intersections, or a loss of mesh quality. Topological changes, such as the merging of neighboring ice structures or the formation of highly concave regions, are also difficult to handle robustly using an explicit surface representation. Although several successful treatments have been proposed in the literature [72], maintaining a valid surface and volume mesh remains a challenging aspect of multilayer simulations involving complex ice shapes.

Interface Capturing

Interface-capturing methods represent the interface implicitly through a field variable defined over the surrounding computational domain. In the level-set method, an n-dimensional interface is embedded within an (n+1)-dimensional space and tracked through the evolution of a scalar function ϕ defined in that higher-dimensional domain [73]. The ice–air interface is then recovered as the zero level set of ϕ:

$$\Gamma = \{\boldsymbol{x} \,|\, \phi(\boldsymbol{x}) = 0\} \,. \tag{2.18}$$

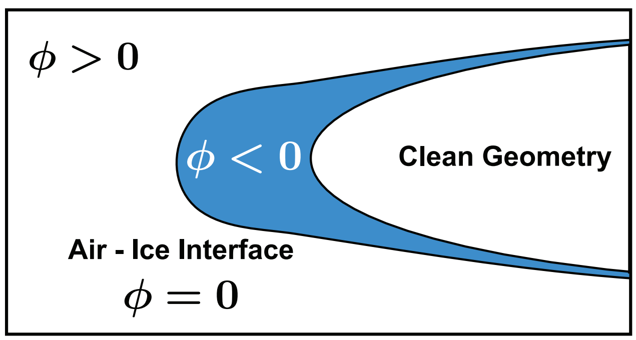


Figure 2.6 Representation of the level-set function ϕ. The ice region is defined by $\phi < 0$ (blue), while the surrounding air corresponds to $\phi > 0$. The interface is implicitly represented by the zero level set $\phi = 0$.

As illustrated in Figure 2.6, the level-set function is commonly defined as positive in the fluid region, negative within the solid, and zero at the interface. Its evolution is governed by an advection equation of the form

$$\frac{\partial \phi}{\partial t} + \boldsymbol{V} \cdot \nabla \phi = 0 \tag{2.19}$$

where $\boldsymbol{V}$ is an extension of the interface-growth velocity to the computational region surrounding the interface. When ϕ is maintained as a signed-distance function, the interface-normal vector and curvature can be evaluated as

$$\boldsymbol{n} = \frac{\nabla \phi}{|\nabla \phi|}, \tag{2.20}$$

and

$$\kappa = \nabla \cdot \left(\frac{\nabla \phi}{|\nabla \phi|} \right). \tag{2.21}$$

Several attempts have been made to introduce level-set representations into aircraft ice-accretion simulations. Peña et al. [24] presented one of the first applications of a level-set method to the evolution of an ice shape. Their work demonstrated the feasibility of representing ice growth implicitly, but the application was limited to single-step accretion. Beaugendre et al. [74] employed a level-set representation in a different icing context, namely to describe the evolving geometry and trajectories of ice fragments during shedding simulations. In this case, the implicit representation facilitated the treatment of moving solid interfaces and their interaction with the surrounding flow.

Bourgault-Côté et al. [2] and Lavoie et al. [25] subsequently extended the use of level-set methods to multilayer ice-accretion simulations. Their studies considered both body-fitted and immersed-boundary frameworks.

An alternative formulation was proposed by Donizetti et al. [75], who introduced a discrete level-set representation constructed from a Lagrangian description of the surface. In this approach, the interface remains explicitly represented, while the level-set values at surrounding mesh points are recovered geometrically from the distance to the Lagrangian surface. The level-set field is therefore used primarily as an implicit geometric description rather than as the solution of a continuous advection equation.

Level-set methods offer several advantages for ice-accretion simulations. They can accommodate large interface displacements and topological changes without requiring the connectivity of an explicit surface mesh to remain unchanged. These properties make them attractive for the development of robust three-dimensional multilayer geometry-evolution procedures. However, the signed-distance property:

$$|\nabla \phi| = 1 \tag{2.22}$$

is generally not preserved as the interface evolves. A reinitialization or redistancing procedure is therefore commonly required to recover a suitable distance field [76]. This step must be performed carefully, since it may displace the interface, alter the enclosed ice volume, or smooth sharp geometric features. The updated surface must then be extracted from the implicit level-set representation before the surrounding volume mesh can be regenerated. Particular care is required during this surface-reconstruction and volume-remeshing process to preserve distinctive features of the original clean geometry, especially in regions unaffected by ice accretion, while accurately representing the newly formed ice structures.

2.6 Geometry Representation

The robustness of aircraft icing simulations is strongly influenced by the geometry-evolution and mesh-update procedures. In conventional body-fitted approaches, the computational mesh must conform to the aircraft surface and must therefore be deformed or regenerated after each accretion layer. As the ice shape develops, the surface may exhibit sharp gradients, horns, scallops, and other localized irregularities that make it increasingly difficult to preserve high-quality near-wall elements and avoid mesh degeneration or failure.

Immersed-boundary methods provide an alternative framework in which the solid geometry is represented independently of the underlying computational mesh. The origins of the immersed-boundary approach are commonly traced to the pioneering work of Charles Peskin [77], who introduced forcing terms into the Navier–Stokes equations to represent the interaction between blood flow and the moving boundaries of heart valves. Since this early contribution, a wide range of immersed-boundary formulations has been developed.

In an immersed-boundary framework, the physical boundary is generally allowed to intersect the computational cells rather than coincide with mesh faces. The boundary conditions may then be enforced through modifications of the governing equations, the numerical fluxes, or the discrete solution near the interface. Because the mesh does not need to conform explicitly to the surface, these methods can substantially simplify grid generation and geometry updates.

Immersed-boundary methods have gained considerable attention and have been applied to a broad range of problems, including inviscid flows [78, 79], turbulent flows [80, 81], multiphase flows [82, 83], and fluid–structure interaction [84–86]. More recently, they have also been introduced into aircraft ice-accretion simulations [25, 40].

Their use in aircraft icing, however, remains comparatively limited. In particular, immersed-boundary approaches were reported by only one participant in each of the first two Ice Prediction Workshops, indicating that their application to complete ice-accretion frameworks is still at an early stage of development [22, 23]. Additional work is therefore required to assess their accuracy, robustness, and compatibility within a coupled multilayer ice-accretion framework.

Following the classification proposed by Sotiropoulos and Yang [87], immersed-boundary methods may be divided into three broad families: geometrical methods, continuous-forcing methods, and discrete-forcing methods. Geometrical methods modify the computational mesh, control volumes, or numerical integration procedure in the vicinity of the immersed interface. Continuous-forcing methods introduce forcing or source terms into the governing equations to represent the effect of the solid boundary. Discrete-forcing methods instead impose the boundary conditions directly within the discretized equations, typically by reconstructing or modifying the solution near the interface. The following sections summarize the main characteristics of these three approaches.

2.6.1 Geometrical Methods

The principal geometrical approach is the cut-cell method. In this formulation, the solid interface is represented by geometrically modifying the cells of a background mesh that it intersects. The portions of these cells located inside the solid are removed, producing control volumes that conform locally to the immersed boundary while leaving the remainder of the mesh unchanged. One of the earliest cut-cell formulations was introduced by Clarke et al. [3] for Euler-flow simulations around multi-element airfoils. A general illustration of the cut-cell approach is provided in Figure2.7.

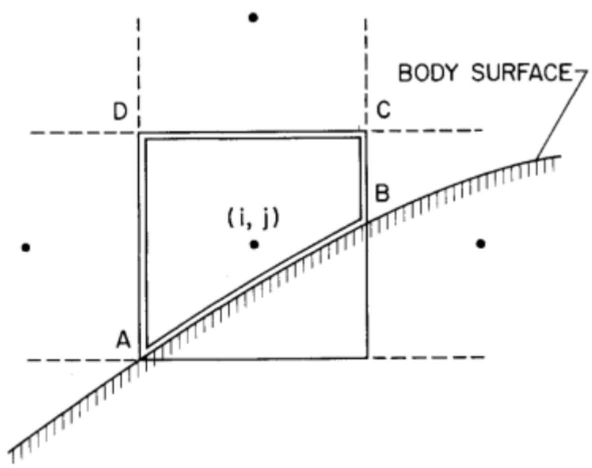


Figure 2.7 Illustration of the boundary cut-cell method adapted from Clarke et al. [3].

Because the interface is represented through an actual modification of the computational cells, some authors treat cut-cell methods separately and do not classify them strictly as immersed-boundary methods.

Cut-cell methods provide a sharp representation of the interface and allow boundary conditions and numerical fluxes to be imposed directly on the reconstructed faces. They are also naturally suited to conservative finite-volume formulations, although the solver must support general polygonal or polyhedral control volumes.

Their main limitations are the complexity of the geometric preprocessing, particularly in three dimensions, and the formation of very small cut cells when the interface passes close to a mesh node or face. These cells may impose severe time-step restrictions in explicit schemes and degrade numerical stability. To alleviate these problems, several remedies have been proposed in the literature, including cell merging [88], cell linking [89], and conservative cell mixing [90].

2.6.2 Continuous-Forcing Methods

Continuous-forcing methods impose the immersed-boundary conditions by introducing a distributed force directly into the continuous governing equations. For example, the incompressible Navier–Stokes equations may be written as

$$\nabla \cdot \boldsymbol{u} = 0 \tag{2.23a}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u} + \boldsymbol{f} \tag{2.23b}$$

where $\boldsymbol{u}$ is the fluid velocity, ρ is the density, p is the pressure, ν is the kinematic viscosity, and $\boldsymbol{f}$ is the forcing term used to represent the immersed boundary. The expression of $\boldsymbol{f}$ distinguishes the different continuous-forcing formulations.

The original immersed-boundary method was introduced by Peskin for the simulation of blood flow interacting with elastic heart structures [77]. The fluid equations are solved on an Eulerian background mesh, whereas the immersed boundary is represented by massless Lagrangian markers with coordinates $\boldsymbol{X}(s,t)$, where s denotes the curvilinear coordinate along the immersed boundary and t is time. The force exerted by the boundary on the fluid is distributed to the surrounding Eulerian points through a regularized Dirac function:

$$\boldsymbol{f}(\boldsymbol{x},t) = \int_\Gamma \boldsymbol{F}(s,t)\delta(\boldsymbol{x} - \boldsymbol{X}(s,t))\ \mathrm{d}s \tag{2.24}$$

where $\boldsymbol{F}$ is the force defined on the Lagrangian boundary Γ. Conversely, the velocity of the markers is interpolated from the Eulerian velocity field according to

$$\frac{\partial \boldsymbol{X}(s,t)}{\partial t} = \int_\Omega \boldsymbol{u}(\boldsymbol{x},t)\delta(\boldsymbol{x} - \boldsymbol{X}(s,t))\ \mathrm{d}\boldsymbol{x} \tag{2.25}$$

This two-way transfer allows the fluid to displace the immersed boundary while the boundary exerts a force on the fluid. The regularized Dirac function spreads this interaction over several grid points, resulting in a diffuse rather than sharp representation of the interface. Although this formulation is well suited to deformable interfaces, its extension to rigid bodies is more problematic. As the boundary stiffness is increased to suppress deformation, the constitutive laws used to determine the restoring force become increasingly stiff and may become ill posed in the rigid-body limit [91].

These limitations motivated the development of penalization methods for rigid boundaries. Initially introduced by Arquis and Caltagirone to model flow through porous media [92], the method was subsequently extended to represent solid obstacles as regions of very low permeability [93]. A mask function distinguishes the solid and fluid domains:

$$\chi(\boldsymbol{x}, t) = \begin{cases} 1, & \boldsymbol{x} \in \Omega_s \\ 0, & \boldsymbol{x} \in \Omega_f \end{cases} \tag{2.26}$$

where Ω_s and Ω_f denote the solid and fluid regions, respectively. For the incompressible Navier–Stokes equations, the no-slip condition may be imposed through the penalization force

$$\boldsymbol{f} = \frac{\chi}{\eta} \left(\boldsymbol{u}_s - \boldsymbol{u} \right) \tag{2.27}$$

where $\boldsymbol{u}_s$ is the prescribed solid velocity and η is a permeability or penalization parameter. In the fluid region, $\chi = 0$ and the original equations are recovered. Within the solid, $\chi = 1$, and a sufficiently small value of η drives the fluid velocity toward the prescribed solid velocity.

Penalization methods require only limited modifications to an existing solver, with the main preprocessing operation consisting of identifying the solid region and constructing the mask function. They also possess a well-established mathematical foundation and can be combined with a wide range of spatial discretizations [94]. However, the boundary is generally represented diffusely, which commonly limits the classical formulation to first-order spatial accuracy near the interface. Higher-order corrections have nevertheless been proposed [95]. Finally, the penalization parameter must be selected carefully: Decreasing η improves the enforcement of the boundary condition but increases the stiffness of the governing equations and may therefore require careful numerical treatment. Although the example presented above involves a single penalization term, the enforcement of other boundary conditions may require several distinct penalization parameters. Their simultaneous selection can introduce additional case-dependent tuning and may require manual adjustment.

2.6.3 Discrete-Forcing Methods

Discrete-forcing methods impose the immersed-boundary conditions directly within the discretized governing equations. Their development is commonly traced to the work of Mohd-Yusof [96], who modified the discrete momentum equations near the immersed interface to enforce a prescribed velocity. This idea was subsequently formalized and extended by Fadlun et al. [97].

For illustration, the incompressible momentum equation may be written as

$$\frac{\partial \boldsymbol{u}}{\partial t} = \boldsymbol{R} + \boldsymbol{f} \tag{2.28}$$

where $\boldsymbol{R}$ contains the convective, pressure, and viscous contributions,

$$\boldsymbol{R} = -\boldsymbol{u} \cdot \nabla \boldsymbol{u} - \nabla p + \frac{1}{Re}\nabla^2 \boldsymbol{u} \tag{2.29}$$

and $\boldsymbol{f}$ denotes the discrete forcing. Using a forward-Euler time discretization gives

$$\frac{\boldsymbol{u}^{n+1} - \boldsymbol{u}^n}{\Delta t} = \boldsymbol{R}^n + \boldsymbol{f}^n \tag{2.30}$$

Imposing a target velocity $\boldsymbol{u}_{\text{target}}$ at the forcing point leads to

$$\boldsymbol{f}^n = -\boldsymbol{R}^n + \frac{\boldsymbol{u}_{\text{target}} - \boldsymbol{u}^n}{\Delta t} \tag{2.31}$$

Although this formulation is expressed through a forcing term, it does not generally need to be evaluated as an independent physical quantity. Instead, the velocity update is directly constructed so that the desired target value is obtained.

The accuracy of the method depends primarily on the definition of $\boldsymbol{u}_{\text{target}}$. The boundary value may be imposed directly from the nearest computational point, resulting in a staircase representation of the interface. Alternatively, the target value may be weighted using the local solid volume fraction to obtain a smoother transition. More accurate approaches reconstruct the target value from neighboring fluid data through interpolation [98]. Unlike continuous-forcing methods, discrete forcing does not introduce a large source term into the continuous equations and therefore avoids the associated stiffness. However, its implementation depends on the spatial and temporal discretizations and generally requires geometrical preprocessing near the interface.

A widely used form of discrete forcing is the ghost-cell immersed-boundary method, developed notably by Tseng and Ferziger [4]. It generalizes the ghost-cell treatment of body-fitted boundaries to interfaces that do not coincide with the computational mesh. Cells located inside the solid but required to complete the stencil of neighboring fluid cells are identified as ghost cells. For each ghost cell, a wall point is determined on the immersed interface and an image point is constructed in the fluid, commonly along the local wall-normal direction. Since the image point does not generally coincide with a computational node, its solution is interpolated from neighboring fluid cells. The ghost-cell value is then assigned such that interpolation between the ghost and image points satisfies the prescribed condition at the wall.

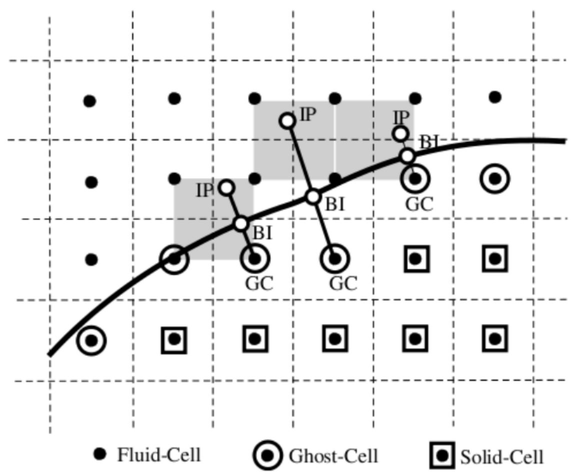


Figure 2.8 Image-point and ghost-cell construction near an immersed boundary, adapted from Tseng and Ferziger [4].

Depending on the interpolation procedure, ghost-cell methods can retain the order of accuracy of the underlying numerical scheme and provide a sharp representation of the immersed interface [4, 99]. The governing equations and flux calculations in the fluid domain remain largely unchanged, with the principal additions consisting of the geometrical preprocessing and the interpolation used to update the image and ghost values. These operations must be performed efficiently to limit the computational overhead.

The main difficulties arise from the construction of suitable ghost, wall, and image points in regions containing high curvature, sharp features, or closely spaced boundaries. The image point may fall inside the solid or may not have a sufficient number of neighboring fluid cells for the required interpolation stencil. The number and placement of ghost-cell layers also depend on the numerical scheme and the width of its computational stencil. These issues become particularly important for complex iced geometries.

In addition to ghost-cell methods, other discrete-forcing formulations have been developed. Capizzano et al. [100, 101] proposed a face-forcing approach in which the boundary condition is imposed through the fluxes of the cell faces intersected by the immersed interface rather than through reconstructed cell values. The procedure remains similar to the ghost-cell approach: a wall-normal direction is constructed through the intersected face, an image point is defined in the fluid, and the state at this point is interpolated from neighboring cells. The reconstructed state is then used to determine the flux at the immersed-boundary face. This approach has been applied to Euler flows and RANS simulations [100, 101].

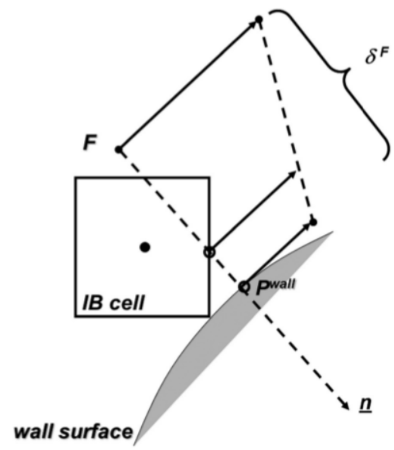


Figure 2.9 Illustration of face forcing near an immersed boundary, adapted from Capizzano et al. [5].

Face forcing avoids some of the difficulties associated with placing several rows of ghost cells near complex interfaces. However, it requires a dedicated treatment of the numerical fluxes on immersed-boundary faces and therefore introduces more extensive modifications to the flow solver than conventional ghost-cell methods.

2.7 Synthesis and Positioning of the Present Work

Despite the continued improvement of the individual modules involved in numerical ice-accretion prediction, the automation and robustness of complete multilayer frameworks remain limited by three closely connected aspects: droplet-impingement modeling, geometry evolution, and the representation of the evolving geometry within the computational mesh. These components are particularly important in three-dimensional simulations, where accurately representing the droplet trajectories and their interaction with increasingly complex iced geometries becomes more challenging, while maintaining a valid surface and a high-quality volume mesh becomes progressively more difficult as the ice shape develops.

Regarding droplet impingement, the literature does not identify a universally superior formulation. Eulerian and Lagrangian approaches provide comparable access to the collection-efficiency distribution, but each presents distinct advantages and limitations. Eulerian methods integrate naturally within finite-volume CFD solvers, are readily parallelized, and remain attractive for large three-dimensional calculations. Their continuum description, however, introduces numerical diffusion near impingement limits and does not directly provide individual impact quantities. Lagrangian methods offer a more direct physical representation of droplet motion and naturally provide impact locations, velocities, and angles. These properties are particularly valuable for SLD modeling and for tracking droplets generated by splashing. Their main limitation is computational cost, especially since a large number of trajectories must be tracked to obtain sufficiently resolved impingement maps. Consequently, the choice between Eulerian and Lagrangian formulations remains dependent on the intended application rather than on a clear overall advantage of either approach.

The literature on geometry evolution reveals a growing interest in interface-capturing methods. Classical interface-tracking approaches remain widely used because of their simplicity and because the evolving surface is available explicitly. Nevertheless, their robustness deteriorates when large displacements, concave regions, sharp features, or surface self-intersections develop. These difficulties become increasingly restrictive in three-dimensional multilayer simulations, where maintaining a valid surface connectivity is considerably more involved than in two dimensions. Interface-capturing techniques, particularly level-set methods, provide a more flexible alternative because they represent the interface implicitly and can accommodate large deformations and topological changes without explicitly preserving the original surface connectivity. They are therefore well suited to the formation of complex ice morphologies. Their benefits are accompanied by challenges related to signed-distance preservation, preservation of sharp features, and the extraction of an explicit surface required for the subsequent accretion layer.

Geometry representation and volume-mesh generation constitute the third central difficulty. In conventional body-fitted frameworks, the volume mesh must be regenerated or deformed after each accretion step. Although this treatment provides a direct and accurate application of the wall boundary conditions, its automation becomes increasingly difficult as the ice shape develops sharp horns, scallops, and highly concave regions. Immersed-boundary methods have therefore gained considerable attention because they decouple the evolving surface from the underlying volume mesh and can reduce, or in some cases eliminate, repeated body-fitted remeshing.

No single immersed-boundary formulation is equally suited to all the equations involved in an ice-accretion framework. Penalization methods require relatively limited modifications to an existing solver and have proved particularly attractive for Eulerian droplet transport [40,102]. Their extension to aerodynamic simulations involving high-Reynolds-number turbulent flows is less straightforward.

Discrete approaches, such as ghost-cell immersed-boundary methods, provide a sharper representation of the interface and can retain the formal accuracy of the underlying numerical scheme. They are therefore generally better suited to aerodynamic-flow calculations, including compressible and turbulent formulations [40]. Their implementation, however, requires more elaborate geometrical preprocessing and robust interpolation procedures. In particular, the construction of ghost, wall, and image points becomes challenging near sharp ice features, highly curved surfaces, and closely spaced portions of the interface.

The preceding review therefore suggests that the most appropriate methodology for a complete ice-accretion framework may involve different numerical treatments for its individual modules rather than the systematic use of a single formulation. The present thesis follows this direction by focusing on complementary developments for droplet impingement, geometry evolution, and geometry representation. First, a Lagrangian droplet-tracking framework is investigated as an alternative to the established Eulerian formulation, with particular attention to its computational efficiency, collection-efficiency prediction, and ability to represent SLD wall interactions. Second, a level-set method is developed for robust three-dimensional multilayer geometry evolution, with the objective of reducing the dependence of the update procedure on the quality and connectivity of an explicit surface mesh. Finally, the integration of the evolving geometry within an immersed-boundary framework is examined to assess the extent to which repeated body-fitted remeshing can be reduced.

CHAPTER 3 THESIS STRUCTURE

This chapter presents the organization of the thesis and the relationships between its principal scientific contributions. After summarizing the research objectives, it outlines the overall research strategy and briefly introduces the three articles that constitute the main contributions of this work.

3.1 Research Objectives

The general objective of this thesis is to improve the robustness and automation of three-dimensional multilayer aircraft ice-accretion simulations through the development and assessment of numerical methods for droplet impingement, geometry evolution, and geometry representation.

The specific objectives are:

- Develop and assess a Lagrangian framework for droplet-impingement prediction, including Supercooled Large Droplet (SLD) phenomena.
- Develop a robust level-set methodology for three-dimensional multilayer geometry evolution.
- Extend an existing immersed-boundary flow solver to investigate an alternative geometry representation for aircraft ice-accretion simulations.
- Verify and assess the proposed developments using representative analytical, numerical, and experimental icing benchmarks.

3.2 Overall Research Strategy

As described in Chapter 2, aircraft ice-accretion simulations consist of successive aerodynamic, droplet-impingement, thermodynamic, and geometry-evolution calculations. Rather than modifying this overall workflow, the present work develops and evaluates improvements to three components of the numerical framework: droplet impingement, geometry evolution, and geometry representation.

3.3 Presentation of the Articles

This thesis is presented in the form of three scientific articles, each corresponding to one of the principal developments pursued during this research.

3.3.1 Article 1: Lagrangian Droplet Tracking

The first article presents the development of a Lagrangian droplet-impingement framework integrated within the existing icing solver. The work focuses on improving the efficiency and robustness of particle tracking on unstructured meshes while extending the formulation to account for Supercooled Large Droplet effects. The proposed methodology is assessed using representative two- and three-dimensional icing benchmarks.

3.3.2 Article 2: Three-Dimensional Level-Set Geometry Evolution

The second article presents a level-set methodology for three-dimensional multilayer ice accretion. The work focuses on providing a robust geometry-evolution procedure capable of handling successive accretion layers while maintaining compatibility with the remaining components of the numerical framework. The methodology is verified and validated using representative icing configurations.

3.3.3 Article 3: Immersed-Boundary Ice Accretion

The third article investigates the extension of an existing immersed-boundary flow solver to aircraft ice-accretion simulations. The proposed framework integrates aerodynamic flow, droplet impingement, thermodynamic calculations, and geometry evolution in order to evaluate immersed-boundary techniques as an alternative to conventional body-fitted approaches for evolving geometries.

3.4 Relationship Between the Articles

The three articles address complementary aspects of the aircraft ice-accretion process while remaining largely independent. The first article focuses on droplet-impingement prediction, the second on geometry evolution, and the third on geometry representation through immersed-boundary techniques.

Although each development can be incorporated independently into an aircraft icing framework, together they contribute toward a more robust and automated methodology for three-dimensional multilayer aircraft ice-accretion simulations.

CHAPTER 4 Article I: A Ray-Tracing-Enhanced Lagrangian Particle Tracking Framework for Ice Accretion Applications

This paper presents the development of a Lagrangian particle-tracking solver for ice accretion simulations on distributed-memory systems for aircraft applications. The solver is implemented within CHAMPS, an in-house multiphysics framework. The primary aim of this work is to enhance the robustness, accuracy, and computational efficiency of Lagrangian droplet tracking. To this end, a face-to-face tracking scheme is employed for robust boundary-intersection detection, combined with a ray-tracing–enhanced bilinear patch intersection technique to improve performance and scalability. An impact-neighbor identification strategy enables smooth water-catch distributions with a reduced number of droplets, while a hybrid Eulerian– Lagrangian adaptive seeding method automates droplet initialization. Supercooled large droplet (SLD) effects are incorporated to better align predictions with experimental data. A scalable parallel implementation is employed to enable efficient execution on distributed-memory architectures. Results demonstrate favorable verification against Eulerian formulations and improved accuracy relative to experimental data. Compared to existing Lagrangian tracking approaches, the proposed framework provides improved robustness in boundary-intersection detection, increased accuracy in collection efficiency predictions, and reduced computational cost, while maintaining scalability on distributed-memory systems.

4.1 Introduction

Particle-laden flows have long been a subject of scientific investigation due to their prevalence in numerous applications ranging from climate prediction [103] to combustion systems [104] and up to aeronautical applications [48, 105–107]. A key challenge lies in the large number of particles required for high-fidelity simulations, which imposes significant computational demands. With recent advances in high-performance computing (HPC) on distributed systems, there is a growing need to optimize computational frameworks for tracking vast particle populations.

One application of interest for this study is ice accretion in aviation—a phenomenon that occurs when an aircraft traverses a supercooled moist cloud. Ice accretion can severely degrade aerodynamic performance [7,8], making accurate prediction essential for flight safety [108]. The aviation industry increasingly relies on advanced numerical tools to forecast and mitigate the effect of ice formation, with water catch simulations playing a pivotal role in identifying droplet impingement locations.

Recent Ice Prediction Workshops [22, 23] showcase that participants are leaning more toward using Eulerian droplet tracking methods [109–113]. These methods adopt the same computational framework as the flow solver, hence presenting a great advantage in ease of implementation and computational efficiency. However, the Eulerian model still faces challenges in accurately representing complex droplet physics, such as splashing. Conversely, Lagrangian solvers, while more physically representative, have traditionally suffered from low computational efficiency. Nonetheless, modern HPC developments offer a promising pathway for reintroducing Lagrangian approaches in industrial ice accretion prediction tools.

Significant progress has been made in particle-laden flow simulations using the Lagrangian formulation, particularly for ice accretion, with recent studies highlighting strategies to overcome computational challenges. For instance, [106] adapts the grid partitioning scheme based on particle trajectories, while [107] employs a dynamic load-balancing approach in which a dedicated computational rank redistributes droplets across the remaining ranks. However, further work is required to suppress oscillatory behavior [114] not seen in Eulerian results as well as to automate seeding-grid generation and develop computationally efficient techniques that improve accuracy without increasing computational cost. This work addresses several limitations of existing Lagrangian droplet tracking approaches. In particular, a face-to-face tracking strategy is implemented to improve the robustness and efficiency of boundary-intersection detection. To alleviate oscillations and spurious irregularities in the resulting surface-based numerical results, neighboring boundary cells are identified using the gradient stencil, enabling smoother surface results without requiring an excessive number of droplets. A ray-tracing–enhanced bilinear patch intersection technique is further employed to reduce intersection cost and improve solver scalability. In addition, a hybrid Eulerian–Lagrangian adaptive seeding strategy is introduced for large-scale configurations to automate droplet initialization and minimize user intervention. Supercooled large droplet (SLD) physics are incorporated in both the standalone Lagrangian and hybrid Eulerian–Lagrangian solvers to achieve predictions that align more closely with experimental observations. Finally, a hybrid MPI–OpenMP-like parallelization strategy based on Chapel paradigms is employed to enhance computational efficiency on distributed-memory systems. The paper is structured as follows: Section 4.2 reviews droplet tracking techniques in ice accretion applications and introduces the proposed Lagrangian approach. Section 4.3 details the solver enhancements, including ray-tracing–driven face traversal, adaptive seeding, computation of collection efficiency with a gradient-stencil weighting strategy, incorporation of supercooled large droplet (SLD) effects, and the parallelization strategy for improved performance. Section 4.4 presents and analyzes benchmark test cases, highlighting key observations and trends. Finally, Section 5 summarizes the main findings and outlines future research directions.

4.2 Droplet Tracking Techniques

4.2.1 Droplet Tracking Approaches for Ice Accretion

Water droplet tracking is a key component of ice accretion simulations, as it determines both impingement locations and the amount of water collected on solid surfaces. Numerous approaches have been proposed in the literature, all aiming to balance robustness, accuracy, and computational efficiency.

Historically, early ice accretion solvers such as LEWICE [115] relied on Lagrangian tracking, in which individual droplet trajectories are computed by solving their equations of motion. While well suited to panel-based potential-flow solvers, the extension of this approach to modern RANS-based frameworks introduces additional complexity due to the need for repeated interpolation of flow quantities at droplet locations.

To address these limitations, Eulerian approaches [44] were introduced, modeling droplets as a continuous dispersed phase governed by transport equations for momentum and volume fraction α (Fig. 4.1). This formulation integrates naturally within flow solvers and offers excellent computational efficiency and scalability . However, accurately representing complex droplet physics, such as splashing and secondary breakup, remains challenging and often computationally expensive.

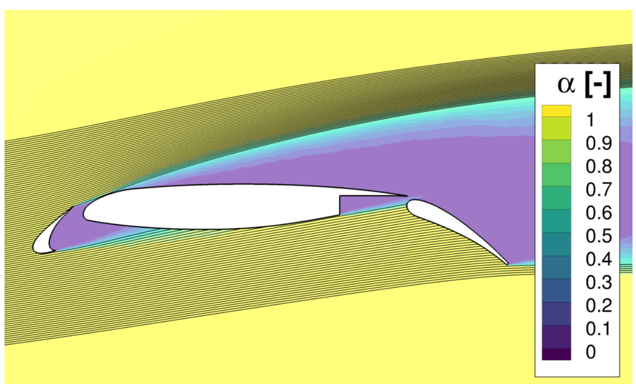


Figure 4.1 Example of Eulerian solver results : Countour of water volume fraction α with droplet trajectories streamlines.

Recent studies have nonetheless highlighted the advantages of Lagrangian tracking for capturing localized phenomena, particularly in the formation of scalloped ice structures [116]. By resolving individual droplet trajectories, the Lagrangian approach can better capture localized enrichment effects and fine-scale surface features that may be smoothed out in Eulerian formulations (e.g. between scallops or small-scale ice features).

4.2.2 Lagrangian Droplet Tracking

Lagrangian droplet tracking is based on integrating the equation of motion from Newton's second law (Eq. 4.1) for each droplet individually. First, the velocity of the droplet $\boldsymbol{u_d}$ is calculated using the air flow velocity $\boldsymbol{u_a}$ and the buoyancy-corrected gravitational acceleration $(\rho_d - \rho_a)/\rho_d\, \boldsymbol{g}$. Then a second integration (Eq. 4.2) gives the position of the droplet $\boldsymbol{x_d}$. For the present work, the forces considered are the drag force as a result of the fluid flow and the gravitational force acting on the droplet. The scalar function for the drag force $\mathbf{F}$ can be found using the droplet local Reynolds number based on the relative velocity $\|\boldsymbol{u_a} - \boldsymbol{u_d}\|$ and droplet diameter D. This leads to a force coefficient F (Eq. 4.3) using C_D, the drag coefficient of the droplet, ρ_a, the air density and ρ_w, the water density:

$$\frac{d\boldsymbol{u_d}}{dt} = \frac{\mathbf{f}}{m} + \mathbf{g} = F(\boldsymbol{u_a} - \boldsymbol{u_d}) + \frac{\rho_d - \rho_a}{\rho_d}\mathbf{g} \tag{4.1}$$

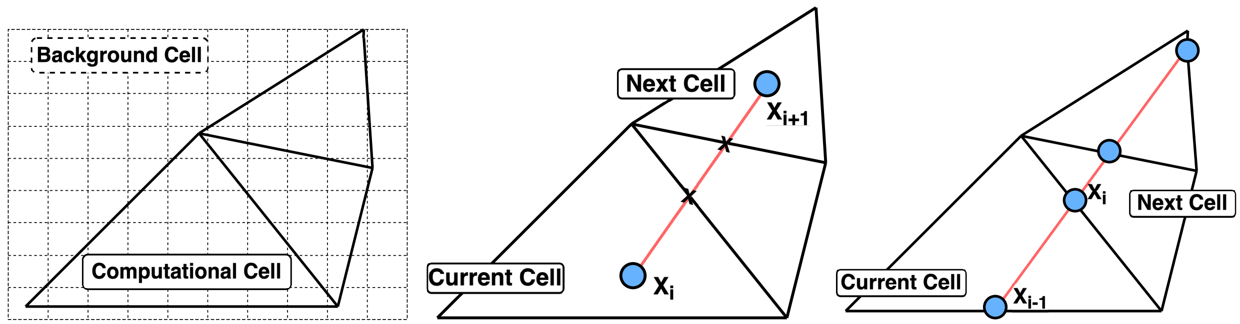
$$\frac{d\boldsymbol{x_d}}{dt} = \boldsymbol{u_d} \tag{4.2}$$

$$F = \frac{3C_D\rho_a\|\boldsymbol{u_a} - \boldsymbol{u_d}\|}{4\rho_w D} \tag{4.3}$$

The key hypotheses to ensure the validity of this model are the following:

- The mass of the droplet is supposed constant and does not change over time before colliding with the geometry;
- The droplets are independent from one another, therefore collisions and coalescence are not considered;
- The droplet concentration do not affect the fluid flow;
- The effects of the turbulence are neglected on the droplets;
- The droplets do not rotate or have lift.
- The phase change within the droplets is not considered, as the solver is limited to computing the deposited liquid water mass under the quasi-steady icing hypothesis [117].

One of the main problems of Lagrangian droplets trackers in such applications is that droplet trajectories rarely coincide with the flow grid. The integration of Eq. 4.1 relies on interpolation of the flow velocity depending on the droplet position. Therefore, after each integration, a relocating step is needed in order to pinpoint the exact location of the droplet inside the flow grid. The choice of the technique is based on accuracy, efficiency and stability. In the literature, various techniques have been developed for particle tracking. Tree-based search techniques [118] and more advanced auxiliary-grid approaches [119], illustrated in Fig. 4.2a, provide a straightforward way to identify host cells for tracked droplets. However, both approaches incur significant computational and memory overheads and are difficult to deploy efficiently on distributed-memory systems.



(a) Auxiliary grid tracking [119]. (b) face-to-face tracking [48] (In Cell Positions). (c) face-to-face tracking [105] (On Face Positions).

Figure 4.2 Droplet tracking techniques.

Face-to-face searching techniques [48], illustrated in Fig. 4.2b, exploit the droplet's previous position and velocity to locate its host cell by traversing mesh connectivity. These methods can become challenging to handle when droplets traverse near-wall zones, particularly in regions with strong gradients, where careful treatment and appropriate time-step selection are required to ensure robust impact detection.

The face-to-face tracking method presented by [105] further refines this process using edge-to-cell (in 2D) or face-to-cell (in 3D) connectivity, as illustrated in Fig. 4.2c. Similarly to the previous method, a ray is casted based on the droplet velocity. Then the algorithm iteratively computes the nearest face intersection to find the next flow cell. This approach offers significant flexibility since the timestep becomes an adaptive by-product of the simulation. Specifically, a local timestep is calculated based on the droplet's residence time within each cell. This adaptability is important in Reynolds-Averaged Navier–Stokes (RANS) simulations, where fine meshes are needed to resolve near-wall flows and the resulting small cells demand reduced timesteps for accuracy and stability. An additional advantage of this method is its simplicity of implementation, as it leverages the same finite-volume framework used by the flow solver. Accordingly, the Lagrangian droplet tracking is performed on the same computational domain and unstructured mesh as the flow simulations. Moreover, the host cell is implicitly determined through mesh connectivity, eliminating the need for explicit host cell searches. The inherent alignment with the finite-volume solver ensures compatibility and seamless integration. These advantages collectively make it the preferred choice for particle tracking in this work.

The integration of Eq. 4.1 is achieved using a second order backward difference scheme [105]:

$$a\boldsymbol{u_d}^n + b\boldsymbol{u_d}^{n-1} + c\boldsymbol{u_d}^{n-2} = F(\boldsymbol{u_a} - \boldsymbol{u_d}) + \mathbf{g} \tag{4.4}$$

$$\boldsymbol{u_d}^n = \frac{F\boldsymbol{u_a} - b\boldsymbol{u_d}^{n-1} - c\boldsymbol{u_d}^{n-2} + \mathbf{g}}{a + F} \tag{4.5}$$

$$a = \frac{2\Delta t_c + \Delta t_{c-1}}{(\Delta t_c + \Delta t_{c-1})\Delta t_c} \tag{4.6}$$

$$b = \frac{-(\Delta t_c + \Delta t_{c-1})}{\Delta t_c \Delta t_{c-1}} \tag{4.7}$$

$$c = \frac{\Delta t_c}{(\Delta t_c + \Delta t_{c-1})\Delta t_{c-1}} \tag{4.8}$$

where Δt is the time step at the current iteration. The superscripts n, $n-1$, and $n-2$ represent time integration levels, while the subscripts c and $c+1$ denote cell values. The algorithm presented in [105] consists of an iterative process that ends up converging towards the droplet velocity. The flow velocity at the face $\boldsymbol{u_a}^n$ is calculated through averaging flow velocity across current and the next cell:

$$\boldsymbol{u_a}^n = \frac{1}{2}(\boldsymbol{u_{a}}_c + \boldsymbol{u_{a}}_{c+1}) \tag{4.9}$$

This presumes that the next cell is known in advance, hence the iterative process. At first, $\boldsymbol{u_d}^n$ is supposed to be equal to $\boldsymbol{u_d}^{n-1}$. Then, the nearest face intersection is computed on the basis of that assumption. This leads to calculating the time step along with the other local variables using Eqs. 4.6, 4.7, and 4.8. Finally, Eq. 4.5 is then used to calculate $\boldsymbol{u_d}^n$ and the process is iterated until there is a slight user specified difference between two consecutive iterations. It should be noted that the closest face intersection is made based on the mid-cell velocity :

$$\boldsymbol{u_d}^{n-\frac{1}{2}} = \frac{1}{2}(\boldsymbol{u_d}^n + \boldsymbol{u_d}^{n-1}) \tag{4.10}$$

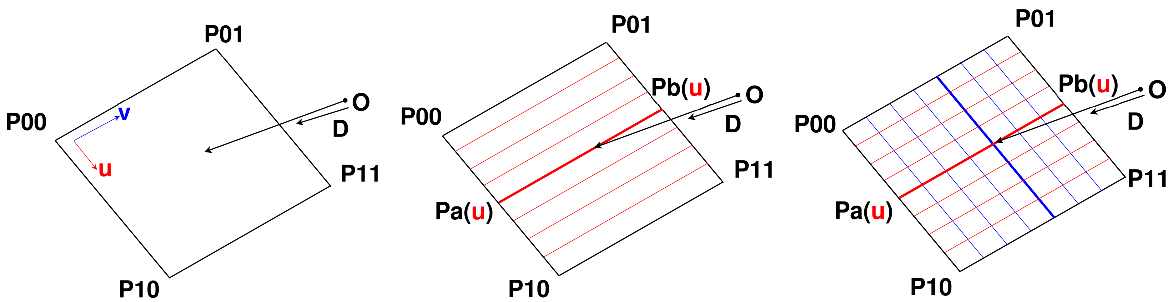
Due to high gradients near the wall, this method can reduce to the first order [105]. To alleviate this problem, the flow gradient is used to interpolate the flow velocity from the center of the cell to the droplet position using the distance $\boldsymbol{d}$ between the two:

$$\boldsymbol{u_a}_{face} = \boldsymbol{u_a}_{cell} + \nabla \cdot \boldsymbol{d} \tag{4.11}$$

4.3 Improvements in Lagrangian Particle Tracking

4.3.1 Ray–Quadrilateral Intersections in CFD Applications

In computational fluid dynamics (CFD), determining the intersection between particle velocity rays and mesh elements is a critical step in computing particle trajectories. When meshes are composed of quadrilateral elements, intersection detection becomes notably more challenging. Quadrilateral faces are not guaranteed to be perfectly planar due to numerical discretization errors. This lack of coplanarity complicates the definition of a unique intersection plane, making classical planar ray–plane tests unreliable.



(a) Step 0: Definition of the ray and bilinear patch geometry.
(b) Step 1: Determination of the intersection u-parameter along the bilinear patch.
(c) Step 2: Determination of the intersection v-parameter along the bilinear patch.

Figure 4.3 Steps for Ray–Bilinear Patch Intersection.

A commonly used workaround is to split each quadrilateral into two planar triangles and perform independent ray–triangle intersection tests, each solved following the method of [120]. Although straightforward, this approach introduces artificial discontinuities along the internal diagonal, potentially causing small numerical inconsistencies and reduced computational efficiency. A more consistent alternative is to represent each quadrilateral as a bilinear patch, parameterized in local coordinates $(u, v) \in [0, 1]^2$, as illustrated in Fig.4.3a. In this parametric representation, the surface is defined by bilinear interpolation of the four corner vertices, providing a continuous geometric mapping over the quadrilateral. This formulation allows intersections to be computed directly on the bilinear surface, maintaining geometric continuity across the element patch.

Recent advances in computer graphics have introduced highly optimized algorithms for ray–patch intersection. Among these, the GARP (Geometric Approach to Ray–Patch Intersections) method [121] offers a robust and efficient alternative to the algebraic formulation presented in [122]. It is this latter formulation that remains widely used in Lagrangian particle-tracking frameworks [123, 124].

Ray–patch intersection aims to determine where a ray $\boldsymbol{r(t)}$ (Eq.4.12), emitted from a given position, intersects a patch $\boldsymbol{p(u, v)}$ represented in parametric coordinates (Eq.4.13):

$$\boldsymbol{r}(t) = \boldsymbol{O} + t\boldsymbol{D}, \quad t \geq 0, \tag{4.12}$$

$$\boldsymbol{p}(u, v) = (1-u)(1-v)\boldsymbol{p}_{00} + (1-u)v\boldsymbol{p}_{01} + u(1-v)\boldsymbol{p}_{10} + uv\boldsymbol{p}_{11}. \tag{4.13}$$

where $\boldsymbol{O}$ and $\boldsymbol{D}$ denote the ray origin and direction, respectively, and $\boldsymbol{p}_{ij}$ are the patch vertices. Equating Eqs. 4.12 and 4.13 yields a system of three equations that can be solved analytically [122]. While this approach is conceptually simple and easy to implement, it introduces algebraic branching and conditional checks that can hinder computational performance. Instead of explicitly computing an intersection plane or solving high-order polynomial systems, the GARP method interprets the bilinear patch as a ruled surface obtained by linear interpolation between two opposite edges. The intersection procedure is carried out in two parametric stages, as illustrated in Fig. 4.3b–4.3c.

First, the intersection is searched along the u-parametric direction. For a given ray, the parameter u is determined by enforcing coplanarity between the ray and the interpolated patch edge. This condition is expressed by setting the scalar triple product of the vectors $\boldsymbol{OP}_a$, $\boldsymbol{P}_a\boldsymbol{P}_b$, and $\boldsymbol{D}$ to zero, which leads to a quadratic equation in u. Only solutions satisfying $u \in [0, 1]$ are retained.

Once u is determined, the corresponding interpolated segment $\boldsymbol{P}_a\boldsymbol{P}_b$ is fully defined. The second stage consists of determining the v-parametric coordinate by projecting the ray onto this segment and identifying the point that minimizes the distance between the ray and the segment. A valid intersection is obtained when $v \in [0, 1]$, after which the ray parameter

t is computed. In the presence of multiple valid solutions, the intersection corresponding to the smallest positive value of t is selected. The reader is referred to [121] for a detailed derivation of the algorithm, along with a discussion of numerical robustness and performance considerations.

To evaluate computational performance, one million random rays and quadrilateral patches were generated for intersection testing using both formulations, along with the traditional two-triangle method. Results demonstrate that both ray–patch methods outperform the two-triangle approach, with the GARP method achieving an additional 15% speed-up over the analytic formulation of [122]. The implications of these improvements on the overall solver scalability are discussed in Section 4.4.

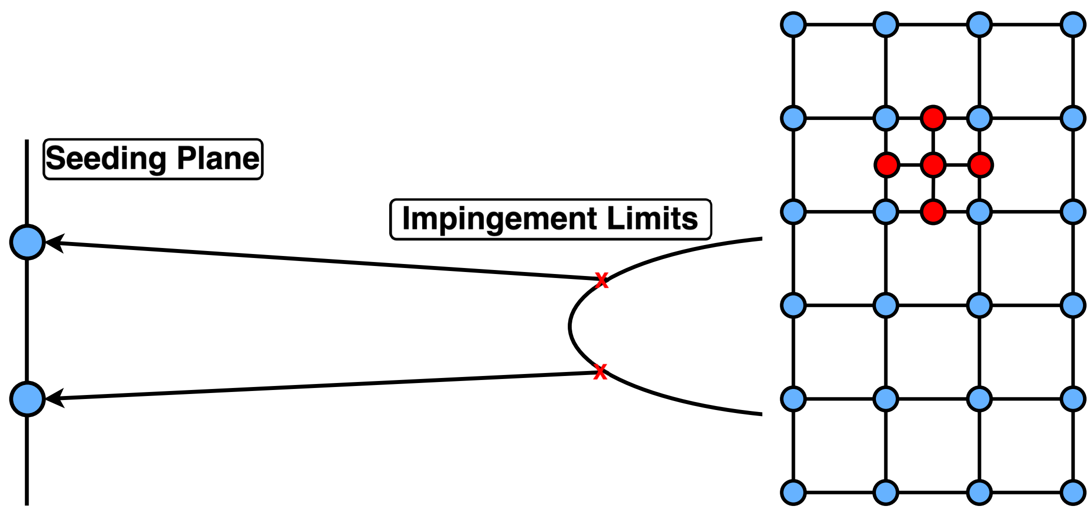
Table 4.1 Performance comparison of ray–quadrilateral intersection methods.

Intersection Method	**Speed-up vs. Two-Triangles**	**Speed-up vs. [122]**
Ramsey et al. [122]	51.3%	0 %
GARP [121]	58.2%	14.2%

4.3.2 Adaptive Seeding

In the Lagrangian droplet tracking framework, the strategic placement of the seeding grid plays a key role throughout the process. Typical seeding techniques [125] involve projecting the surface of interest onto a user-specified seeding plane and refining the grid to achieve optimal results. For instance, the process often begins with a simple rectangular grid, followed by launching a set of droplet trajectories. During tracking, the minimal wall distance is stored, a value typically accessible in RANS simulations. Cells whose minimum wall distance falls below a user-specified threshold, e.g., based on a reference length, are selectively refined to increase resolution in near-wall regions. While this approach is practical and straightforward to implement, careful consideration must be given to the projection of the surface. This is because the projection depends not only on the flow velocity direction but also on droplet diameter, which must account for gravitational effects. As a result, the placement of the seeding grid requires significant user expertise, especially when dealing with a wide range of flow conditions and droplet sizes.

To address these challenges and further automate the process, this work builds on the two-dimensional hybrid method proposed by [126], with the seeding technique is extended to three-dimensional simulations and parallelized. The approach traces droplet trajectories backward, starting from the surface of the geometry and extending upstream to a user-defined seeding plane. However, initiating this process within a Lagrangian solver poses challenges because the wall boundary condition in RANS simulations imposes zero velocity at the surface. To address this, the droplet velocity field obtained from an Eulerian solver (previously implemented in CHAMPS [127]) offers a practical solution, as droplets possess velocity before impacting the surface. The suggested method employs a hybrid Eulerian-Lagrangian seeding technique (illustrated in Fig. 4.4a), where an Eulerian solution is used to prepare the seeding grid for the Lagrangian solver.



(a) Hybrid seeding [126]. (b) Refinement technique.

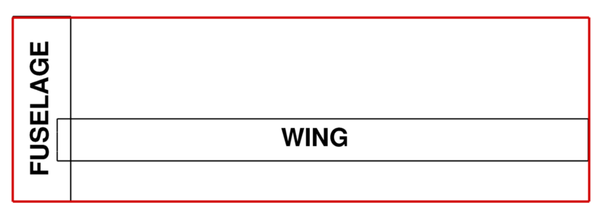
Figure 4.4 Adaptive seeding.

For the present work, the hybrid Eulerian-Lagrangian seeding technique is adopted. The Eulerian velocity field is used to perform reverse trajectory computations. Face intersections are computed based on the inverse droplet velocity, which is derived from the Eulerian solution. While Eulerian results are generally sufficient, Lagrangian results provide a more accurate representation of droplet physics, as will be demonstrated in Section 4.4. The process continues until the trajectory intersects a user-specified upstream seeding plane. To account for potential dissipation-induced errors, a safety margin can be incorporated into the computed seeding grid position. This automated approach significantly simplifies the process, reducing user input to specifying only the x-coordinate for the starting position. Once the limits are defined, a structured 2D rectangular grid (or 1D line grid for 2D problems) is generated, with nodes corresponding to the seeding positions.

Another concern inherent in the Lagrangian method is determining the number of droplets used for accurate and smooth solutions. This work integrates an adaptive seeding technique as presented in [48]. Initially, a simulation is conducted with a user-specified number of droplets, and subsequently, the seeding mesh is refined. For instance, if a droplet impacts the geometry, the corresponding seeding cell is then refined. The stop criterion for refinement is based on achieving a small residual for a global deposition metric (for instance, $\epsilon \approx 10^{-5}$). This metric is defined as the surface integral of the corresponding local quantity (Section 4.3.3) over the body, providing a global measure of droplet deposition. After each refinement loop, the older droplets are restarted with the previous solution, hence only newly created droplets are simulated. The enhanced accuracy and smoothness will be discussed in Section 4.4.

While adaptive seeding is an effective technique to achieve mesh convergence with the seeding grid, it requires a substantial number of initial droplets to accurately target complex geometries. In other words, there must be enough droplets impacting the geometry in the first refinement iteration in order to capture the geometry accurately. This can be challenging for large-scale geometries such as case 2.1 from the fifth edition of the High Lift Prediction Workshop (HLPW5) [128] depicted in Fig. 4.5a. The current technique employs structured rectangular meshes based on the bounding box (highlighted in red Fig. 4.5b), whose limits are obtained through backtracked droplets. However, this approach is not ideal for cases with multiple components. The overall bounding box becomes far too large and includes areas where unnecessary droplet trajectories are simulated. To circumvent this issue, a seeding grid per component is employed, as shown in Fig. 4.5b. Each component is simulated and refined individually before adding its contribution to the total results. This approach allows for better targeting each component without substantially increasing the number of droplets to be simulated.

(a) Case 2.1 - HLPW5 geometry.



(b) Large scale geometry seeding grid example.

Figure 4.5 Seeding technique for large scale geometries.

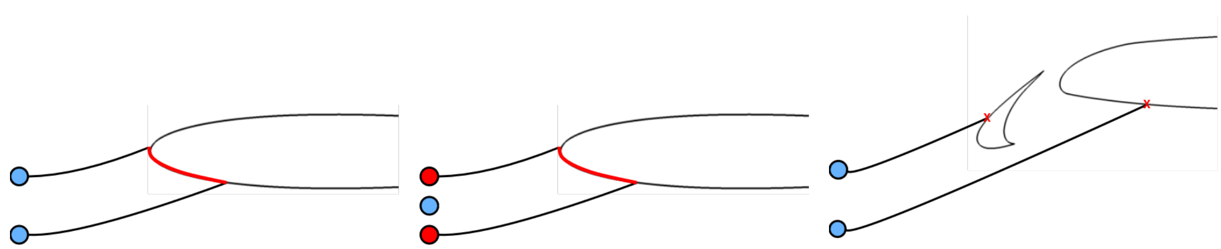
4.3.3 Collection Efficiency Computations

In addition to impingement locations, water droplet tracking outputs the collection efficiency on the geometry. The latter measures how much of the upstream water is eventually captured by the geometry. This metric is essential for the thermodynamic exchanges in the ice accretion process, as it determines the incoming water mass flux on the surface. In the Lagrangian model, the traditional approach constructs a flow tube using three (triangle) or four (quadrilateral) trajectories in 3D, or two (line) trajectories in 2D, and tracks it downstream until it intersects the geometry. The collection efficiency β is then computed as the ratio between the upstream area of the flow tube and the curvilinear area on the surface [129]:

$$\beta = \frac{Upstream\ Area}{Impingement\ Area} = \frac{dy}{ds} \tag{4.14}$$

However, this method encounters several difficulties:

- Inconsistent impingement locations: There is no guarantee that droplets from the same streamtube will impact the same geometric component. For instance, as depicted in Fig. 4.6c, one droplet may impinge on the main element, while another may impacts the flap or slat. Identifying which impingement area to use becomes ambiguous, and in many cases, computing such an area is practically infeasible. To avoid such occurrences, the spacing between droplets in the seeding grid must be kept sufficiently small. This approach inevitably increases the number of droplets to simulate.
- Sparse droplet impacts: An inadequate number of droplets reaching the surface can result in fluctuations in the calculated collection efficiency, ultimately leading to inaccurate predictions of ice shapes.



(a) Flow Tube (b) Local Flow Tube. (c) Pathological case for the collection efficiency.

Figure 4.6 Collection Efficiency computations. Primary and auxiliary droplets are represented in blue and red respectively.

To address these challenges, a technique proposed in [130] introduces the computation of two additional auxiliary trajectories for each droplet , therefore creating a local flow tube for each droplet. The first issue is resolved by releasing the auxiliary trajectories in close proximity to the primary ones by a factor of : $dy \approx O(10^{-8}c_{\text{ref}})$, where c_{ref} is the reference length of the geometry. The latter method increases the likelihood of droplets impacting the same geometric component in complex configurations.

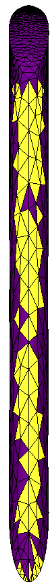
Once all droplet trajectories (both primary and auxiliary) are computed, the corresponding collection efficiencies are determined. The final step is to transfer the local Lagrangian data to the cell-centered framework of the solver. Since multiple droplets can impact the same surface face, and to seamlessly integrate the Lagrangian solver within the CHAMPS framework, a weighted face average is applied during post-processing to assign a single β_i value to the impacted surface cell as presented in Eq. 4.15. The weighting $\omega_{i,d}$ for each droplet is based on the inverse square of the distance between the face center and the droplet's impact position.

$$\beta_i = \frac{\sum \beta_{i,d} \cdot \omega_{i,d}}{\sum \omega_{i,d}} \tag{4.15}$$

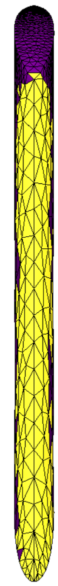
In 3D simulations, efficiently covering all surface faces without significantly increasing the droplet count is challenging. This issue can be alleviated by expanding the influence zone of each droplet. Using the available gradient stencil, neighboring boundary faces can be easily identified and also influenced by a droplet impact, extending its coverage area. Figure 4.7 presents an example of a wind tunnel simulation around an airfoil, highlighting the difference in the impacted area for the same number of simulated droplets. Elements impacted by at least one droplet are shown in yellow (Fig. 4.7a). In Fig. 4.7b, the expanded impact zone is clearly visible, with previously unaffected elements in Fig. 4.7a now included.

4.3.4 SuperCooled Large Droplets Effects Modeling

Supercooled Large Droplet (SLD) studies have gained significant attention following the requirements outlined in Appendix O of the Federal Airworthiness Regulations (FAR). SLD effects, such as droplet splashing and bouncing, might enable droplets to reach areas without anti-icing mechanisms. This can result in the formation of ice ridges, potentially causing flow separation and premature stalling of the aircraft. Another notable effect is droplet deformation due to shear stress in the flow, which alters the droplet's shape, modifies its drag coefficient, and consequently change its trajectory course. The study of SLD ice accretion is challenging on both experimental and numerical fronts. Experimentally, obtaining reli-



(a) Without considering neighboring elements.



(b) With considering neighboring elements.

Figure 4.7 Impact zone effect: Yellow-colored cells indicate at least one droplet impact, while purple-colored cells signify no impact.

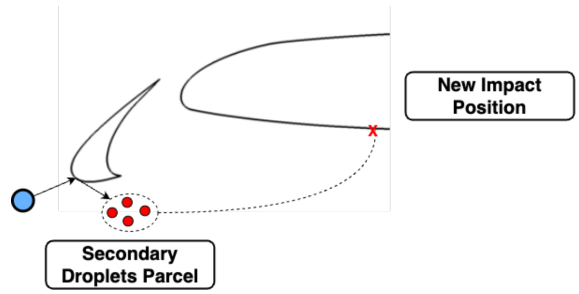
able data is difficult, time-intensive, and expensive. References [131–133] provide extensive datasets on SLD impingement distribution and ice shape characteristics.

SLD behavior differs fundamentally from that of smaller supercooled droplets. Due to their larger diameters ($> 50\mu m$ [134]), SLDs exhibit greater inertia and are less influenced by the flow, resulting in a higher likelihood of surface collision. Additionally, their geometric flexibility and susceptibility to deformation further distinguish them. As SLDs travel downstream, aerodynamic forces induce an uneven pressure distribution across their surface. When these forces exceed the capacity of interfacial and viscous forces to maintain a spherical shape, the droplets undergo deformation or even breakup (Fig. 4.8a). Consequently, droplets transition from a spherical to a nonspherical form, affecting both their trajectory and the volume of deposited water. However, except in freezing drizzle conditions, the impact of these phenomena is generally limited, as noted by [54]. In this study, droplet deformation is considered by adjusting the drag coefficient as proposed by [55], while droplet breakup is not accounted for. The drag coefficient is obtained by blending spherical and oblate-disk correlations as a function of the droplet Reynolds and Weber numbers, as reported in Eq. 4.16 and 4.17. Both $C_{D,\text{sphere}}$ and $C_{D,\text{disk}}$ are functions of the droplet Reynolds number and are taken from [55].

$$C_D = (1 - f)\, C_{D,\text{sphere}} + f\, C_{D,\text{disk}} \tag{4.16}$$

$$f = 1 - \left(1 + 0.07\sqrt{\text{We}}\right)^{-6} \tag{4.17}$$

(a) Droplet Deformation.



(b) Droplet splashing.

Figure 4.8 Droplet Phenomena.

Droplet-wall interactions are particularly complex within the SLD regime. Unlike smaller droplets that tend to fully deposit on the surface, larger droplets often fragment into secondary droplets upon impact, as demonstrated in experiments [135–138] and illustrated in Fig. 4.8b. This fragmentation extends the impingement limits by enabling secondary droplet re-impingement, while lowering the peak collection efficiency. This behavior is influenced by droplet characteristics such as diameter, velocity, impact angle, density, viscosity, and surface tension, which are described using non dimensional numbers like Reynolds, Weber, and Ohnesorge numbers.

Numerically, simulating droplet impacts requires high-resolution meshes to capture interactions at the droplet scale. Methods like Volume of Fluid (VOF) [139] and Moment of Fluid (MOF) [140] have been used for this purpose. However, due to the large number of droplets involved in ice accretion simulations, these methods are computationally prohibitive with current hardware capabilities. Instead, semi-empirical models calibrated against experimental data are commonly employed. These models introduce the mass loss coefficient f_m (or the sticking rate ϵ) to account for the mass loss due to droplet splashing, directly influencing the initially computed collection efficiency $\beta_{i,0}$ as expressed by:

$$\beta_{i,\,final} = (1.0 - f_m)\, \beta_{i,0} \tag{4.18}$$

$$\beta_{i,\,final} = \epsilon\, \beta_{i,\,0} \tag{4.19}$$

Table 4.2 summarizes the splashing models implemented in CHAMPS [141]. Wright et al. [52] enhanced their splashing model by incorporating the impact angle into the splashing parameters and mass loss formula. Their results show that splashing is less likely when droplets impact perpendicular to the wall and more pronounced near impingement limits. The model was calibrated by comparing simulated water collection efficiency with experimental data to ensure accuracy across a wide range of cases. Similarly, Trontin et al. [54] used sub-functions to represent the mass loss rate as a function of incident angle and kinetic energy. By comparing their numerical results with experimental data, they fine-tuned the parameters to improve model precision. Details regarding the models parameters can be found in the Appendix **??**. The effects of both models on the collection efficiency will be examined in greater detail in Section 4.4.

Table 4.2 SLD Splashing Models

Literature Reference	Mass Loss Coefficient
Wright et al. [52]	$0.7(1.0 - sin(\theta_0))[1.0 - e^{-0.0092(K_M-200.0)}]$
Trontin et al. [54]	$1.0 - g(\frac{\theta}{\theta_c})f(\frac{K_c-K_0}{K_0})$

As previously mentioned, depending on the impact conditions, droplet splashing on the surface leads to both mass deposition on the surface and the re-injection of smaller droplets into the flow (Fig. 4.8b). Ongoing studies are currently exploring the dynamics following droplet splashing on the surface, with current models offering limited insight into this phase. For example, [52] provides post-impact variables that enable the simulation of secondary droplets resulting from splashing, as shown in Eq. 4.20 for post-impact velocities and Eq. 4.21 for post-impact diameter. In CHAMPS, once all droplet trajectories are simulated and the sticking rate or mass loss is calculated for each droplet, the secondary droplets are modeled as a single parcel containing the mass from the point of impact. This approach remains a simplification since the SLD models listed in Table 4.2 do not provide detailed information on the nature of splashing. The secondary droplets are injected at the same boundary face where the impacts have occurred.

$$\begin{aligned} u_{n,\,post\,impact} &= u_{n,\,impingement}(1.075 - 0.0025\,\alpha_{impingement}) \\ u_{t,\,post\,impact} &= u_{t,\,impingement}(0.300 - 0.0020\,\alpha_{impingement}) \end{aligned} \tag{4.20}$$

$$\frac{D_{new}}{D} = exp(-0.0281\,K_M) \tag{4.21}$$

The secondary droplets parcel is then tracked until the next impact position. This raises the question of how to properly incorporate the secondary droplets re-impingement contribution into the overall collection efficiency calculation. One approach could be to add the average contribution of all the secondary droplets impacting a given face, as expressed in Eq. 4.22.

$$\beta_{final} = \epsilon\,\beta_{i,0} + \frac{\sum \beta_{i,\,sd} \cdot \omega_i}{\sum \omega_i} \tag{4.22}$$

where ω_i are weights based on the distance between the new position of impact and the center of the corresponding face and $\beta_{i,\,sd} = f_m\,\beta_{i,\,d}$ is the loss in the collection efficiency from the first impact position. Another approach would be to normalize the secondary droplets contribution using Eq. 4.23:

$$\beta_{Re-Impingement} = \frac{m_{sd} \cdot (\boldsymbol{u_{sd}} \cdot \boldsymbol{n})/A_s}{m_\infty \cdot u_\infty / A_\infty} \tag{4.23}$$

where m_{sd} is the mass contained in the secondary droplet parcel, $\boldsymbol{u_{sd}}$ is the parcel velocity at the new impact position, A_∞ is the area of the seeding grid, A_s is the area of the facet on the geometry and m_∞ is the initial injected droplet mass upstream. The latter can be directly added to β_{final} of the boundary face. This latest approach is adopted in CHAMPS as it provides a more natural basis for normalizing the collection efficiency.

4.3.5 Hybrid Eulerian-Lagrangian Solver

Since SLD effects are Lagrangian in nature, they are challenging to implement in an Eulerian solver, particularly when modeling droplet splashing. The initial attempt by [59] involved switching boundary conditions to in-flow boundaries for splashed faces. In their work, a preliminary simulation is carried out to determine initial deposition. Boundary faces identified as splashed were then tagged and clustered based on specific metrics. As dual boundary conditions cannot be applied simultaneously, subsequent simulations were carried out for each cluster of boundary faces. The goal was to treat these faces as in-flow boundaries, injecting the appropriate amount of water droplets into the volume field to account for the re-injection. However, this approach proved computationally expensive, as it required additional simulations for numerous faces. A more recent approach by [142] addressed this issue by simulating all splashed faces simultaneously, assuming that water would not re-impinge on these faces. While this may apply in some cases, it cannot be guaranteed for complex configurations.

A recent technique by [48] involved simulating the secondary droplets from splashing as Lagrangian droplets. This approach combined the efficiency of the Eulerian model for the initial deposition phase with the precision of the Lagrangian solver for tracking secondary droplets, thereby reducing the computational cost associated with simulating a large number of droplets.

The present work builds on this technique to enhance the capabilities of CHAMPS's Eulerian solver. The Eulerian solver is used to compute the initial deposition and apply splashing models to estimate the mass loss. For each splashed boundary face, a parcel of Lagrangian droplets is seeded and tracked using the Lagrangian method. The contribution of re-impingement is accounted by using Eq. 4.22.

4.3.6 Parallelism

The droplet solver in CHAMPS is implemented in Chapel, a high-level open source parallel programming language designed for scalable high performance computing [143]. CHAMPS employs a domain decomposition strategy in which the computational grid is partitioned into mesh partition objects, each corresponding to a spatial subdomain and mapped directly to a computational rank. This design ensures a one-to-one correspondence between mesh partitions and MPI ranks, enabling distributed-memory parallelism while allowing users to adjust the partitioning according to available resources. Load balancing is achieved by partitioning both the flow and droplet seeding grids using the METIS algorithm [144], ensuring that computational work is evenly distributed. Special care is taken to avoid redundant droplet computations at zone interfaces by assigning shared droplets to the lowest zone ID. Intranode communication overhead is minimized via local lightweight mesh copies shared among ranks on the same node, leveraging OpenMP-like parallelism to exploit shared-memory architectures. This hybrid MPI–OpenMP-like approach reduces inter-node communication costs while maintaining scalability across large distributed systems, thus enabling efficient tracking of large droplet populations within the Lagrangian framework.

4.4 Results & Discussions

One of the primary focus of this study lies in the validation of Lagrangian droplet trajectory computations and the SLD implementation, primarily through the computed collection efficiency β. This metric is compared with previously validated Eulerian results in CHAMPS [127] for the droplet trajectory part. Additionally, the impact of SLD effects on the collection efficiency is analyzed using the models previously introduced. The cases presented in Table 4.3 are sourced from both the first and second Ice Prediction Workshop (IPW1-2, [22, 23]) and the fifth edition of the Highlift Prediction Workshop (HLPW5). The Ice Prediction Workshop (IPW) is a series of international initiatives organized by the American Institute of Aeronautics and Astronautics (AIAA) to evaluate and compare the performance of different numerical tools for aircraft icing prediction. It provides experimental datasets covering both 2D airfoils and 3D geometries under various icing conditions. The initial two cases are chosen to assess the implementation on a simplistic airfoil geometry, encompassing both two-dimensional and three-dimensional scenarios. The third case presents a hybrid configuration in a wind tunnel where there is a gap between the top of the airfoil and the tunnel. The final case examines the model performance and accuracy on a larger and more complex scale, with the Common Research Model (CRM). The flow simulations were carried out using CHAMPS RANS solver with Spalart Allmaras (SA) to model the turbulent viscosity. Results were extracted based on information and cut tools provided by the workshop committee when available. Eulerian results were obtained using an Upwind scheme with Van Albada limiter to compute the convective fluxes. Finally, all simulations were carried out using multiple bins distributions. The distributions provided by the IPW1-2 were utilized for the concerned cases. A Freezing Rain (FR) distribution from Appendix 0 is applied for the fourth case to simulate a more realistic scenario for such case. Usually poly-dispersed clouds are chosen due to the absence of a guarantee in experimental results regarding a single diameter size for all droplets. Depending on the experimental setup, a distribution of droplet diameters is provided, indicating the likelihood of each diameter using a cumulative density function. The collection efficiency β_i for each diameter bin is computed sequentially, with each bin assigned a weight γ_i. At the end of the simulation, all contributions are summed using Eq. 4.24:

$$\beta_{i,total} = \sum_{n=1}^{N_{bins}} \beta_i * \gamma_i \tag{4.24}$$

Table 4.3 Cases parameters

CASE	Description	MVD[μm]	AOA[°]	Ma[-]	C_{ref}[inch]	T [K]	Re[-]
IPW1 - 112	NACA64A008 H-Tail	92	6.0	0.23	37.65	280.0	$5 \cdot 10^6$
IPW1 - 122	Three Element Airfoil	92	4.0	0.23	36.0	278.0	$4.8 \cdot 10^6$
IPW2 - 1.3	CRM Hybrid	25	3.7	0.21	39.37	247.15	$8.1 \cdot 10^6$
HLPW5 - 2.1	CRM Wing-Body	FR	6.0	0.20	275.8	288.15	$5.4 \cdot 10^6$

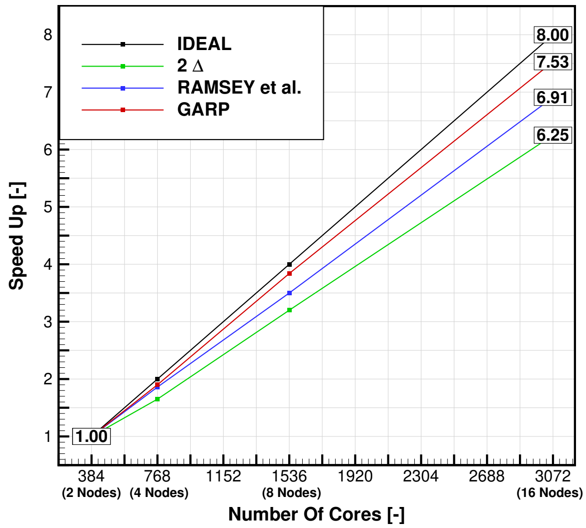
4.4.1 Scalability

To evaluate the efficiency of the parallelization of the Lagrangian solver, Case IPW2 - 1.3 is selected as the benchmark. All the necessary simulations are conducted on the Digital Research Alliance of Canada Trillium cluster. Each node in this cluster is equipped with two sockets, each containing 2 AMD EPYC 9655 (Zen 5, 2.6 GHz), resulting in a total of 192 cores per node. Simulations are conducted using a minimum of 2 and a maximum of 16 nodes, with the node count doubling at each measurement point.

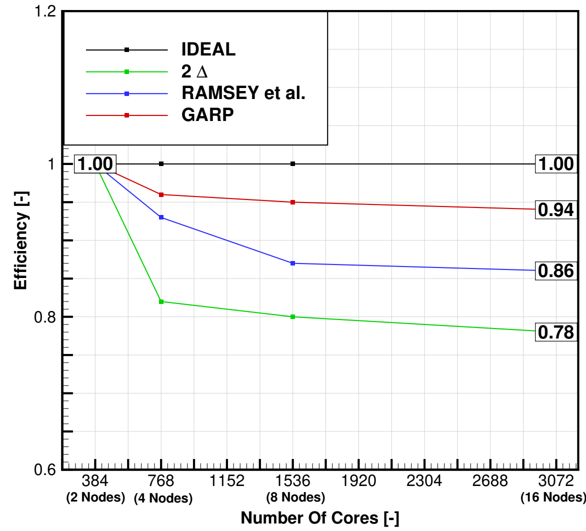
The flow conditions from the original workshop case are adopted. A single droplet diameter, corresponding to the Mean Value Diameter (MVD) of 25 μm, is employed. A total of 12.8 million droplets are seeded for the benchmark. This choice is based on the number of droplets needed for convergence and ensured optimal load balancing across all nodes, facilitating a more accurate comparison of performance.

Unlike Eulerian simulations, where the iteration time correlates to solver efficiency, this metric is not directly applicable to Lagrangian solvers. This is because droplets may require different numbers of iterations to either impact the geometry or move past it. Consequently, the metric selected for comparison is the total computation time.

The computational speedup is determined using Eq. 4.25, and the efficiency is computed using Eq. 4.26. To account for the choice of two nodes as the baseline for comparison, a factor of 2 was included in the efficiency calculation. The results of this study are presented in Fig. 4.9. Results show only a 6% loss in efficiency when using 16 nodes with the GARP technique, which demonstrates better scalability compared to both the method presented in [122] and two-triangle intersection methods. A contributing factor to GARP's improved scalability is that it replaces the two-triangle split with a single, robust ray–patch intersection evaluation, reducing computational overhead and parallel-communication costs. It also involves substantially less branching than the method used in [122], leading to more predictable memory access. Overall, the solver's parallel performance aligns well with trends reported in the literature, confirming its robustness and computational efficiency on distributed-memory



(a) Solver Speed Up.



(b) Solver Efficiency

Figure 4.9 Lagrangian Solver Scalability.

systems.

The loss in efficiency can be associated with differences in droplet trajectories: for instance droplets reaching the leading edge of the geometry follow shorter paths than those reaching the trailing edge. While METIS effectively partitions the seeding grid to optimize load balancing, even small variations in trajectory execution times can degrade the performance of ranks handling droplets that traverse the full geometry. One potential solution is to refine the initial partitioning using ParMETIS [144]. However, this requires precomputing additional droplet trajectories, which increases the overall computational cost.

$$Speed\,Up = \frac{Simulation\,Time\,2\,Nodes}{Simulation\,Time\,N\,Nodes} \tag{4.25}$$

$$\eta = 2 \cdot \frac{Speed\,Up}{N\,Nodes} \tag{4.26}$$

4.4.2 Case IPW1 - 112 : NACA64A008 Horizontal Tail

Figure 4.10 presents impingement results of Case IPW1 - 112. The geometry of the case is provided in Appendix ??. Within Fig. 4.10a, the effects of extending the impact zone of a droplet become apparent. When examining identical droplet quantities, the Lagrangian solution without the extension reveals undesired oscillations, because the present work solves continuous equations, which are not expected to produce oscillations, unlike experimental measurements which contains discrete sets each with their own experimental uncertainty. The newly purposed enhancement gives smooth results, and more importantly the new collection efficiency matches the Eulerian results. Lagrangian trajectories exhibit similar outcomes when compared to Eulerian results for both single diameter (92 μm Fig. 4.10a) and using the diameter distribution provided by the workshop in Fig. 4.10c. The use of diameter distributions affects collection efficiency, as shown in Fig. 4.10b. Results using a diameter distribution exhibit extended impingement limits (approximately 0.1 in difference in the surface distance on the lower surface), due to smaller droplets following the flow streamlines more closely. These droplets are less likely to impact near the stagnation point but have a higher probability of colliding farther downstream.

The second objective to this study is to outline the SLD outcomes on the aforementioned cases (Table 4.3). This serves the dual purpose of validating the implementation of the SLD models and examining the impacts of integrating such variables into consideration. The results consist of comparing the distribution β for the previously presented SLD models in both formulations. Employing these approaches has a significant impact on the computed collection efficiency, by bringing its amplitude and limits closer to the experimental data. This is illustrated in Fig. 4.10d when comparing with the experimental results of [133]. Without considering the effects of SLD, the computed collection efficiency is significantly higher than the available experimental data. The model provided by [52] (denoted as "MODEL-L") is specifically engineered to minimize the SLD effects near the stagnation point and to gradually increase these effects when moving towards the impingement limits. This characteristic can be seen in its formulation: as the impact angle approaches 90° in this region, the mass loss coefficient is approximately zero, indicating that no splashing occurs when the impact angle is normal to the surface. However, further down the geometry, the mass loss effects become more apparent since the impact angle becomes more tangential to the surface. This results in the same solution as the one without SLD effects at the stagnation point, with a gradual decrease in β thereafter.

On the other hand, the model provided by [54] (denoted as "MODEL-O") operates differently. This model assumes that near the stagnation point, when the impact angles are normal to the surface, the droplets should exhibit mass loss depending on their impact velocity. Higher velocities increase the likelihood of droplet splashing, thereby increasing the mass loss coefficient in this region. This approach provides a closer alignment with experimental data as shown in Fig. 4.10d. Both models exhibit similar behavior as they move farther from the stagnation point, tending to accurately predict the impingement limits. CHAMPS results are also compared with the SLD data reported in [48], which employs the same SLD model as in [52] ("MODEL-L"). CHAMPS shows closer agreement with experimental observations, although a direct comparison is difficult given differences in flow conditions, mesh resolution, and other factors. A key distinction lies in the computation of the collection efficiency. In [48], a statistical approach (Eq. 4.27) comparing the upstream water mass flux ($m_\infty u_\infty / A_\infty$) with the local surface flux ($\boldsymbol{u_d} \cdot \boldsymbol{n}/A_s$) is used:

$$\beta = \frac{\sum m_d \cdot (\boldsymbol{u_d} \cdot \boldsymbol{n})/A_s}{m_\infty u_\infty / A_\infty} \tag{4.27}$$

However, this method requires a significantly larger number of droplets to achieve comparable accuracy. In the present work, the local flow-tube technique is employed, yielding accurate estimates of the collection efficiency.

4.4.3 Case IPW1 - 122 : Three Element Airfoil

Case IPW1 - 122 presents a configuration that is particularly challenging because of the re-circulation cove behind the Main element (Fig. 4.11b). Disparities emerge notably in the Main and Flap components of the geometry as was shown by most IPW1 participants (Fig. 4.12a - 4.12b). Fig. 4.13a, 4.13b and 4.13c show the results obtained in CHAMPS for both models using the diameter distribution provided by the workshop. The two models exhibit a good level of agreement. The Eulerian model tends to detect impingement towards the end of the main airfoil component on both sides of the geometry. This discrepancy arises from non-physical values observed for the droplet volume fraction, attributed to dispersion-induced oscillations. The prominence of the Slat component in collecting incoming droplets is apparent, due to its position at the forefront of the geometry. This trend is particularly pronounced on the pressure side, where the geometry is angled at a 4° angle of attack, resulting in a greater accumulation of droplets. This configuration also influences the Main and Flap components, with minimal impingement observed on their suction side. Overall similar results can be observed between the two models. The slight difference between Eulerian and Lagrangian results can be related to dissipation from the Eulerian model, especially near the cove behind the main component as noted in [46]. Ideally, these differences would diminish with mesh refinement as showcased in Appendix **??**.

The SLD results are illustrated in Fig. 4.13d, 4.13e and 4.13f. On the slat component of the geometry, the computations accurately predict the experimental collection efficiency, both in terms of peak values and impingement limits. On the main component, the collection efficiency is significantly underestimated. On the flap part of the geometry, the numerical simulations align more closely with the experimental results. The impingement limits are well predicted. However, the peak value for β is underestimated for the same reason as the main component. When considering secondary droplet re-impingement using the method presented in Section 4.3.4 (denoted “MODEL-L w/ REIMP.”) on both the main component and the flap, an increase in the peak value of β is observed, leading to a slightly better agreement with experimental data.

A similar trend is observed for the hybrid model (denoted “EUL. MODEL-L w/ REIMP. HYBRID”). The effect of re-injected Lagrangian droplets is particularly pronounced on the flap component, where the peak value increases compared to the baseline Eulerian model (denoted “EUL. W/O SLD”). A similar trend is observed across participants’ data and even in more recent results for this case [48, 145]. This raises questions about the reliability of the re-injection model in predicting post-impact variables or/and the validity of the experimental results for this specific case.

4.4.4 Case IPW2 - 1.3 : Hybrid CRM (With Gap)

This case poses significant challenges due to the gap between the top of the airfoil and the wind tunnel wall (Fig. 4.14b). In one of the available meshes from the workshop, the gap was included and explicitly meshed, whereas in the other, the gap was ignored. The presence of this gap presents difficulties at both the flow solver and droplet solver levels, primarily due to the recirculation zone formed in this region, similar to the previous case but here investigated for a 3D geometry. While ice accretion diminishes this gap over time, as demonstrated by experimental ice shapes, the initial layer of ice remains influenced by its presence. Moreover, this test case serves as an excellent opportunity to evaluate the capabilities of the Lagrangian solver under such conditions. The results for this test case are illustrated in Fig. 4.15. The initial comparisons show strong agreement between Eulerian and Lagrangian results accross all the cuts, with discrepancies appearing near the impingement limits. This trend is further highlighted in the contour plot of the collection efficiency shown in Fig. 4.15d. The latter showcase the relative difference between Lagrangian and Eulerian results with respect to collection efficiency calculated as follows:

$$\beta_{\text{RelativeDifference}} = \frac{|\beta_{\text{Lagrangian}} - \beta_{\text{Eulerian}}|}{\beta_{\text{Lagrangian}}} \tag{4.28}$$

Values below 10% are omitted from the plot for clarity (colored in Grey). The observed differences are most pronounced near the impingement region, a consistent trend across all test cases. Additionally, the differences are positive (when not taking the absolute value), indicating that the Lagrangian solver predicts slightly larger impingement limits compared to the Eulerian solver. This observation should be further investigated in future work when comparing ice shapes derived from both models. Other discrepancies are also noticeable near the gap where the Eulerian solver usually have difficulty to predict impingement data.

4.4.5 Case HLPW5 - 2.1 : CRM Wing-Body

The final case, using the Common Research Model (CRM), assesses the solver's performance and accuracy on a larger, more complex configuration representative of an industrial aircraft. The impingement results for Case HLPW5 - 2.1 are presented in Fig. 4.16. These results were extracted using cuts along the wing span and fuselage. The placement of these cuts is detailed in Fig. 4.17.

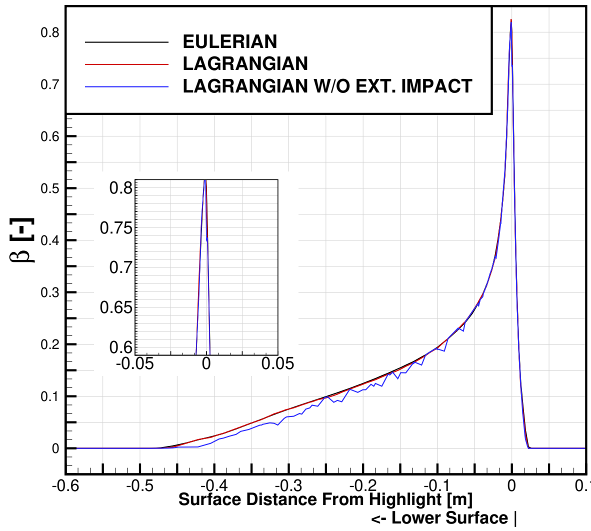
For the fuselage, as shown in Fig. 4.16a, the peak collection efficiency is observed at the nose, gradually decreasing along the surface. The collection efficiency distribution on the wing, depicted in Fig. 4.16b - 4.16f exhibits a non-symmetrical shape due to the 6° angle of attack imposed by the simulation conditions. Higher values of β can be observed at the junction between the fuselage and the wing indicating higher impingement rate in that region. As in previous cases, more droplets are collected on the pressure side of the wing. However, since the CRM model features a swept wing, the effective angle of attack varies along the wing span. This variation is evident when comparing β results on different hingewise cuts along the wing span. Unlike the inboard section (Fig. 4.16b), the third cut shows more droplets collected on the suction side of the wing. The Lagrangian results match with the Eulerian results across all the cuts, ensuring consistency and accuracy in the simulation for such a large scale geometry.

The SLD models implemented in this study exhibit consistent behavior across the full-scale CRM wing-body configuration presented in Fig. 4.16g - 4.16l. Significant mass loss due to splashing is observed across all the cuts. Although acquiring experimental data for such a configuration is challenging, the current results align well with previous findings, demonstrating the robustness of the computational models. This consistent performance across different scales and configurations underscores the models applicability to real-world aviation scenarios.

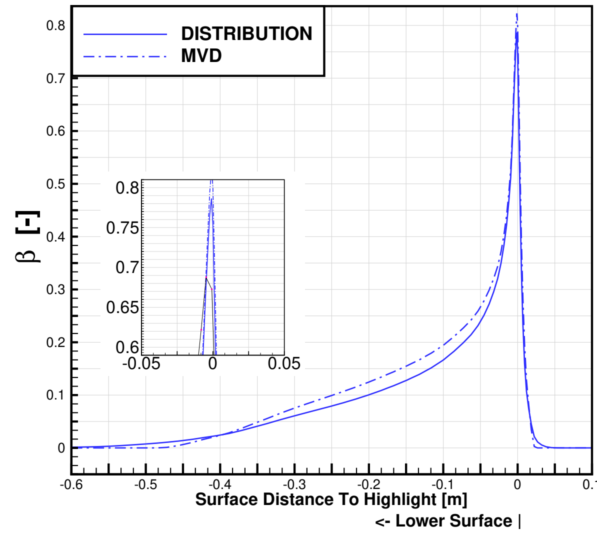
4.5 Conclusion

This work presents a computationally efficient framework for Lagrangian particle tracking, featuring enhancements that increase the robustness, accuracy, and overall efficiency of the solver. The trajectory-tracking algorithm that leverages mesh connectivity precisely determines droplet velocities and positions. It employs a face-to-face approach that dynamically adjusts the droplet time step based on cell size to reduce numerical instabilities, particularly in regions of strong velocity and pressure gradients near walls. A ray-tracing enhanced intersection technique is incorporated to improve the performance of intersection detection and the solver overall scalability. An adaptive seeding strategy is implemented, utilizing Eulerian droplet data to trace trajectories back to a predefined seeding plane, thereby reducing the number of tracked droplets while maintaining accuracy; this is complemented by adaptive mesh refinement of the seeding grid, driven by the accumulated collection efficiency (β) on the surface. To enhance the accuracy of β evaluation, a weighted interpolation based on the squared inverse distance between droplet impact locations and neighboring face centers is adopted, improving droplet coverage and reducing the need for large number of droplets. The solver also incorporates Supercooled Large Droplet (SLD) physics, accounting for mass loss mechanisms such as bouncing, splashing, and rebounding upon impact, which yields a more realistic collection efficiency and closer agreement with experimental data. Finally, a hybrid Eulerian–Lagrangian strategy is discussed, in which Eulerian simulations are used to model the initial deposition while secondary droplets are tracked with the Lagrangian solver, thereby enabling efficient and accurate treatment of SLD effects and complex droplet behavior.

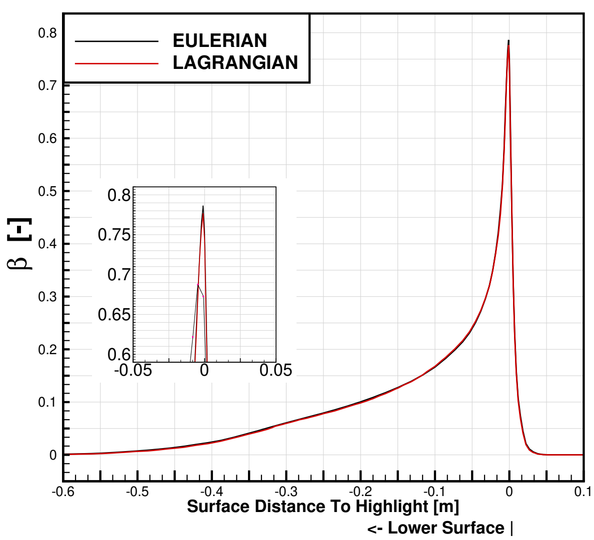
Benchmark cases were selected from the first and second AIAA Ice Prediction Workshops, with a large-scale configuration drawn from the fifth High-Lift Prediction Workshop. The solver achieves 94% parallel efficiency on 16 nodes. Results show strong agreement between the CHAMPS Eulerian solver and the newly developed Lagrangian solver, with SLD simulations aligning more closely with experimental data than other codes reported in the literature. The hybrid Eulerian (for deposition)–Lagrangian(for secondary droplets) approach offers improvements over purely Eulerian predictions, including a slight increase in peak collection efficiency, by combining the computational efficiency of the Eulerian solver for initial deposition with the enhanced droplet physics of the Lagrangian solver for secondary droplets, providing a promising pathway for future developments.



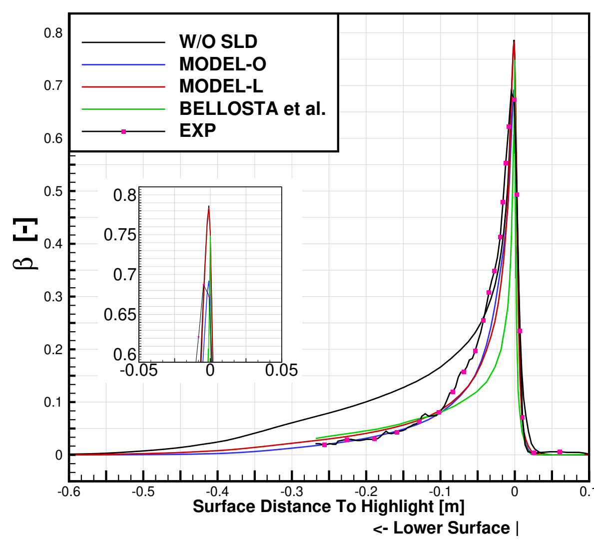
(a) Effect of droplet impact zone.



(b) Effect of droplet distribution on collection efficiency.

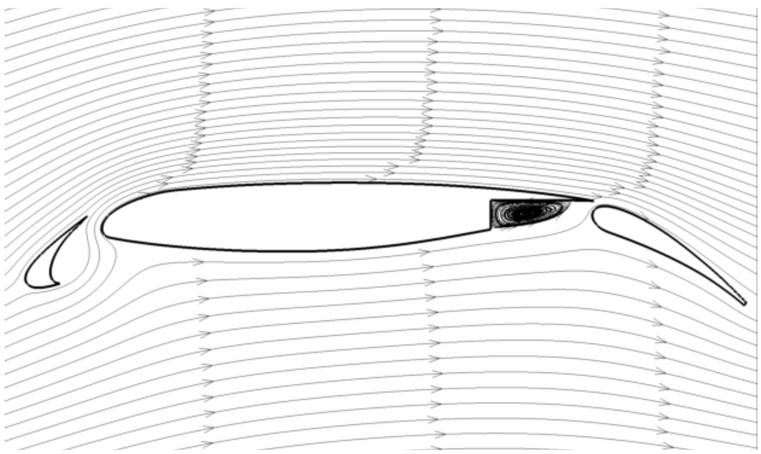


(c) Impingement results.

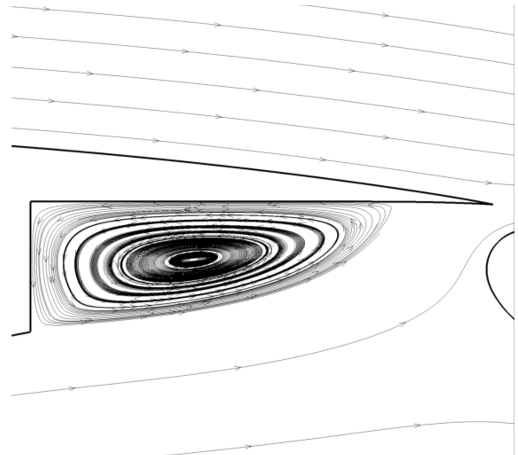


(d) SLD results.

Figure 4.10 Results for CASE IPW1 - 112.

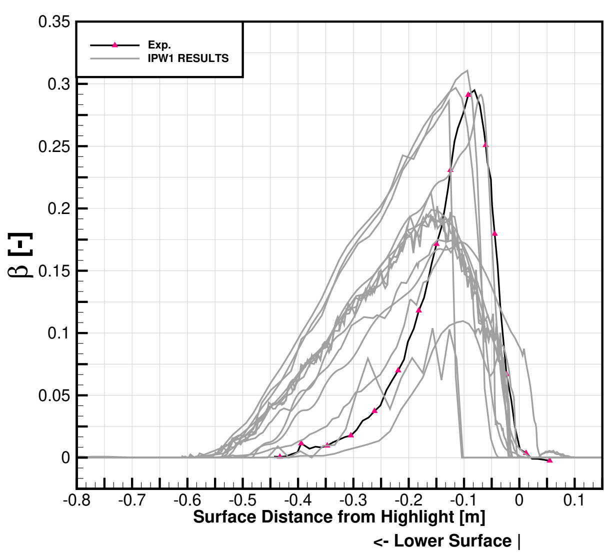


(a) Three Element Airfoil Geometry.

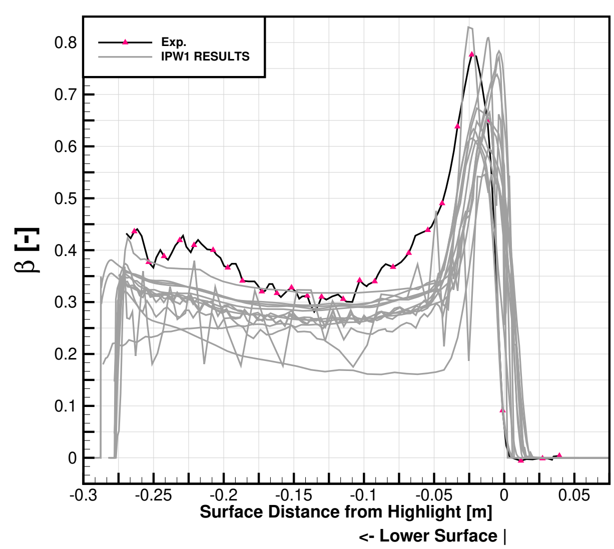


(b) Recirculation zone in the cove of the main element.

Figure 4.11 Description of Case IPW1 - 122.

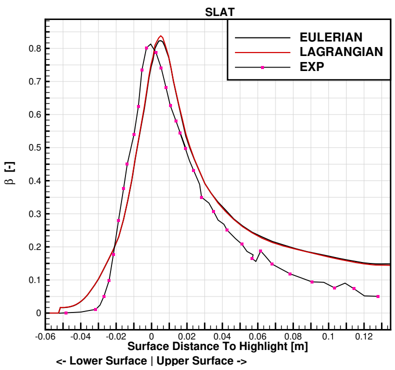


(a) Main component results.

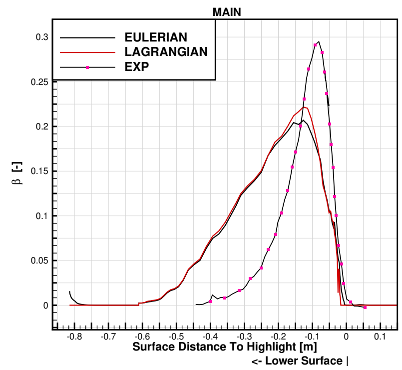


(b) Flap component results.

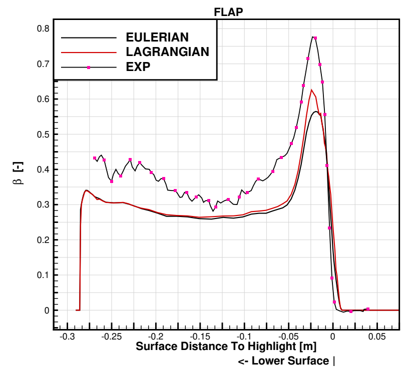
Figure 4.12 Aggregated impingement results of all IPW1 participants for Case IPW1-122.



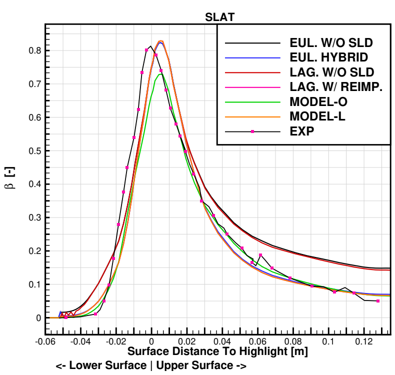
(a) Impingement results on the Slat.



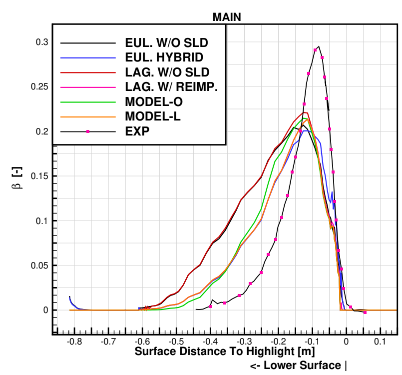
(b) Impingement results on the Main.



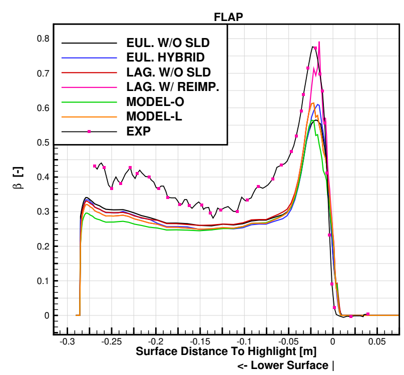
(c) Impingement results on the Flap.



(d) SLD results on the Slat.

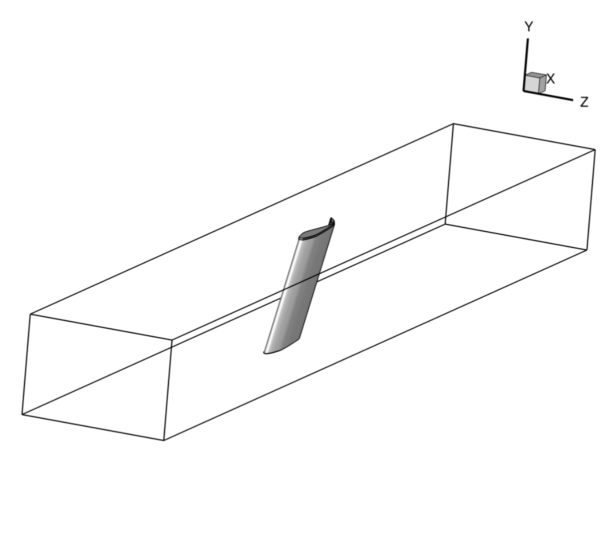


(e) SLD results on the Main.

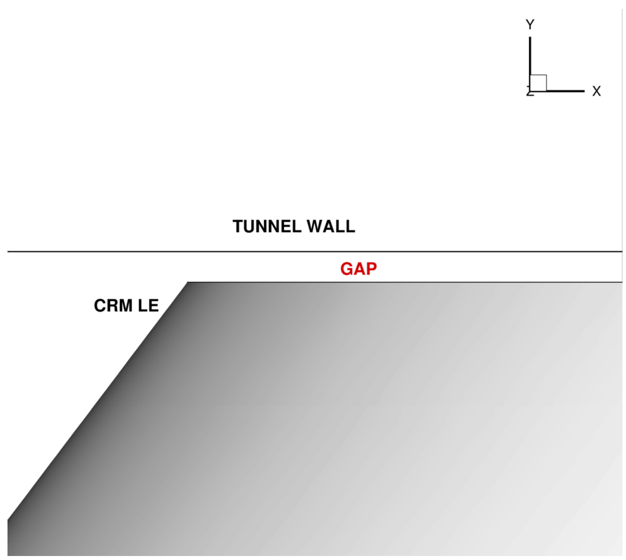


(f) SLD results on the Flap.

Figure 4.13 Results of CASE IPW1 - 122.

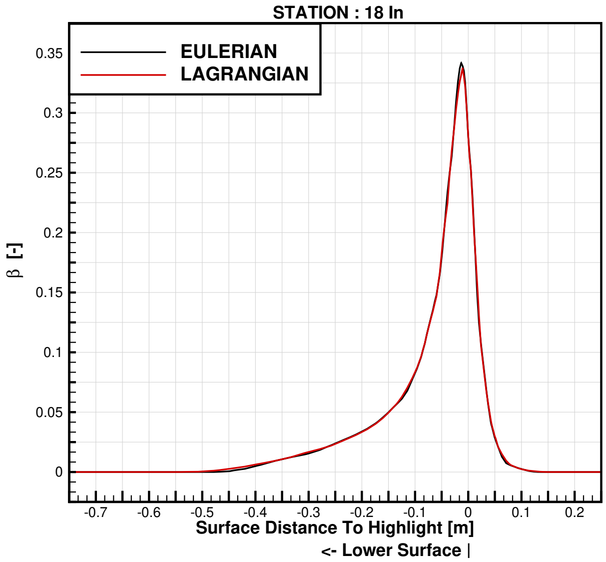


(a) Case IPW2 1.3 Geometry.

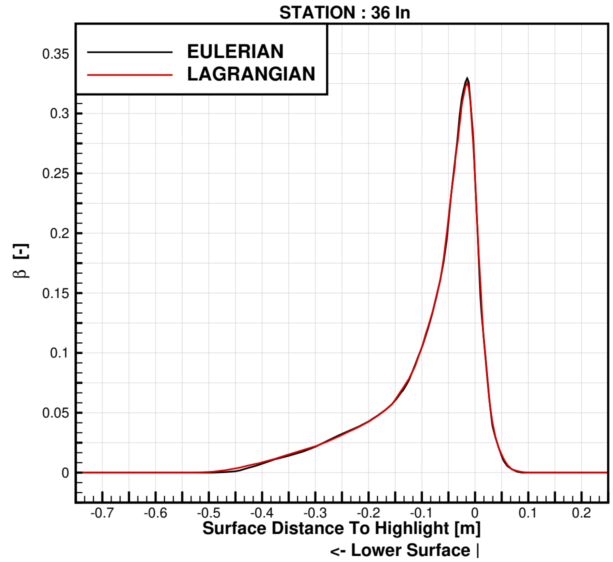


(b) Case IPW2 1.3 Gap region with the wind tunnel.

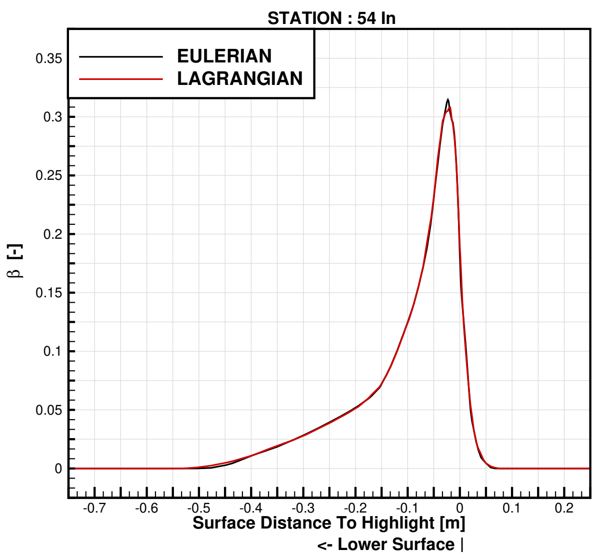
Figure 4.14 Description of Case IPW2 1.3



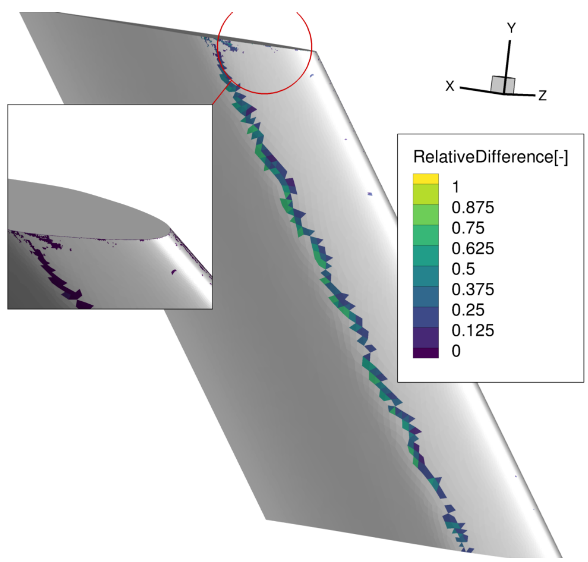
(a) Impingement Results on Station 1 : 18 inches.



(b) Impingement Results on Station 2 : 36 inches.

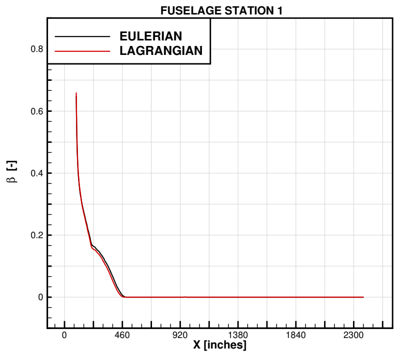


(c) Impingement Results on Station 3 : 54 inches.

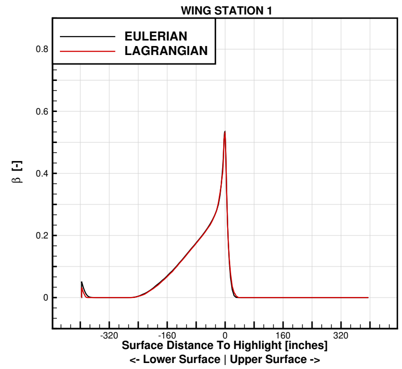


(d) Relative Difference Between Eulerian and Lagrangian Solutions in Terms of Collection Efficiency.

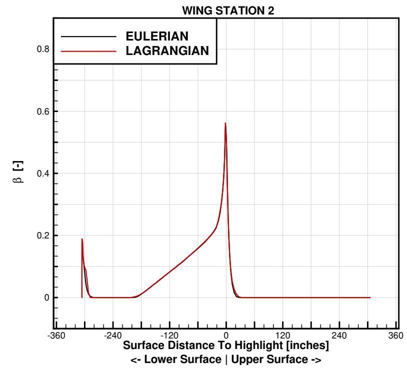
Figure 4.15 Results for CASE IPW2 - 1.3.



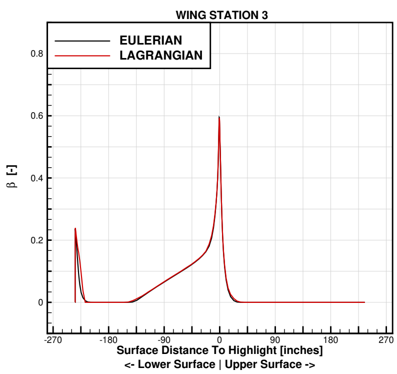
(a) Impingement results on the Fuselage.



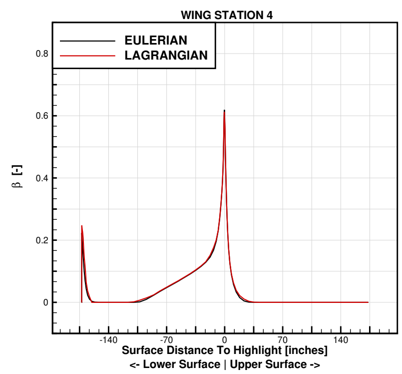
(b) Impingement results on the Wing-1.



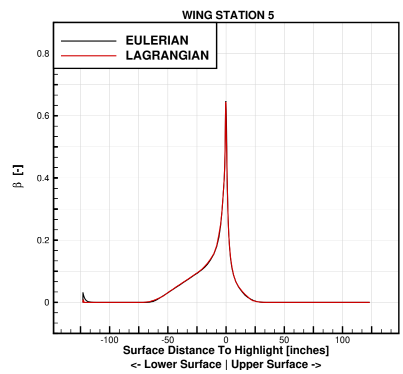
(c) Impingement results on the Wing-2.



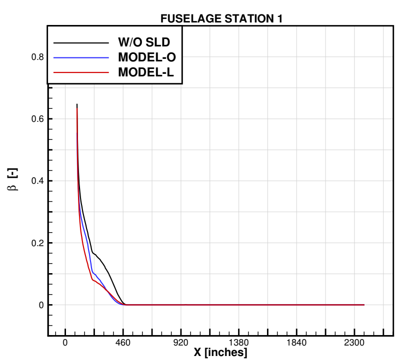
(d) Impingement results on the Wing-3.



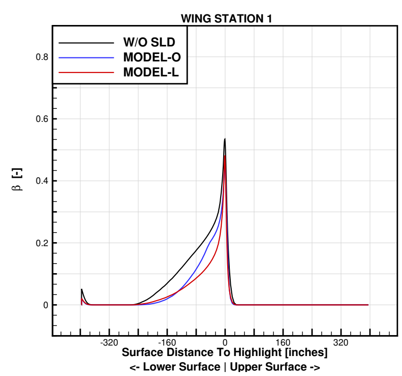
(e) Impingement results on the Wing-4.



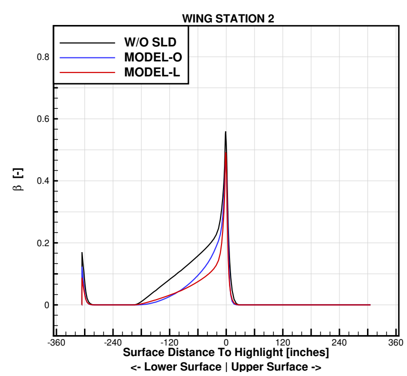
(f) Impingement results on the Wing-5.



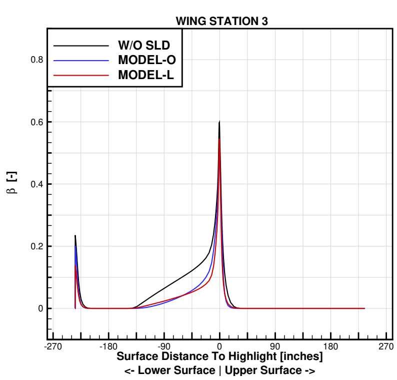
(g) SLD results on the Fuselage.



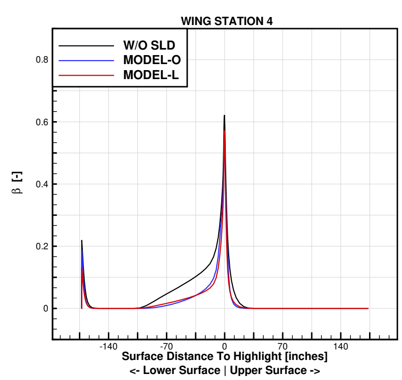
(h) SLD results on the Wing-1.



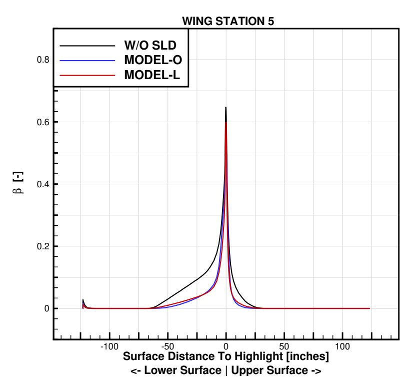
(i) SLD results on the Wing-2.



(j) SLD results on the Wing-3.

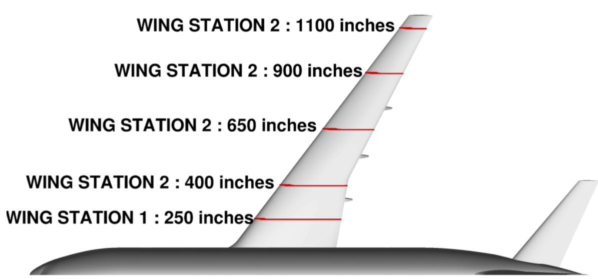


(k) SLD results on the Wing-4.

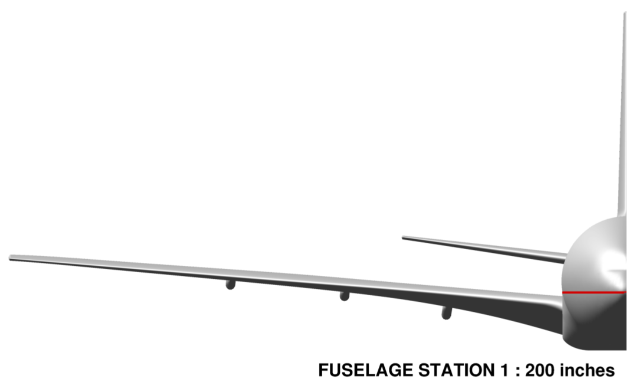


(l) SLD results on the Wing-5.

Figure 4.16 Results for CASE HLPW5 - 2.1.



(a) Collection Efficiency cuts along the wing span.



(b) Collection Efficiency cut on the fuselage.

Figure 4.17 Collection Efficiency cuts.

CHAPTER 5 CONCLUSION

5.1 Summary of Works

5.2 Limitations

5.3 Future Research

Texte / Text.

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APPENDIX A DÉMO

Texte de l'annexe A. Remarquez que la phrase précédente se termine par une lettre majuscule suivie d'un point. On indique explicitement cette situation à LaTeX afin que ce dernier ajuste correctement l'espacement entre le point final de la phrase et le début de la phrase suivante.

APPENDIX B UNE DERNIÈRE ANNEXE / THE LAST APPENDIX

Texte de l'annexe C.